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HOSPITAL AT HOME: A SYSTEMATIC REVIEW OF REVIEWS

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ABSTRACT

Objectives: To provide an overview of the safety and effectiveness of Hospital-at-Home (HaH) according to HaH programme type (early-supported discharge versus admission avoidance), and identify the model which has greater potential to benefit health systems grappling with capacity constraints and rising costs.

Methods: A systematic review of reviews was conducted by performing a search on PubMed, EMBASE, Cochrane Database of Systematic Reviews, Web of Science and Scopus (January 2005 to June 2020) for English-language systematic reviews evaluating HaH. Data on primary outcomes (mortality, readmissions, costs, and length of stay), secondary outcomes (patient/caregiver outcomes), and process indicators were extracted. Quality of the reviews was assessed using AMSTAR-2 checklist.

Results: Ten systematic reviews were identified and classified according to three use cases. Early-supported discharge reviews generally revealed comparable mortality and readmission rates to inpatient care, shorter hospital length of stay, increased total length of stay, and unclear findings for costs. Admission avoidance reviews observed a trend towards lower mortality and costs, comparable or lower readmissions, but unclear findings for length of stay. Among the reviews that included both programme types, reviews specific to chronic obstructive pulmonary disease revealed improved clinical outcomes but unclear findings for resource use.

Conclusion: HaH is a safe alternative to inpatient care and can mitigate some of the challenges facing health systems. Preliminary comparisons suggest that admission avoidance models should be prioritised over early-supported discharge in further explorations of HaH due to potential benefits in costs and clinical outcomes. Further research is needed to clarify outcomes pertaining to costs, caregiver outcomes and adverse events under HaH care.

Protocol registration: There was no registered protocol for this study.

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Strengths and limitations of this study

- The first systematic review of reviews providing a comprehensive overview of the safety and efficacy of HaH
- Highlights evidence gaps pertaining to outcomes and process indicators in the current HaH literature
- Presents a comparison of two major HaH models to offer relevant recommendations for health systems facing capacity constraints and rising costs
- Strength of evidence for comparison between HaH models is low due to heterogeneity in implementation and patient groups across studies

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1. INTRODUCTION

Healthcare systems around the world are grappling with ageing populations, increasing healthcare needs and expenditures, and shortages of hospital beds and manpower[1 2]. Such capacity constraints decrease the efficiency and quality of care delivery, and can lead to delays in elective and emergency treatment[3], increased mortality risk[4 5], lower patient and staff satisfaction, and higher costs for both patients and the health service[6]. Chronic morbidity in ageing populations have further driven the rise in healthcare costs[7]. These complex challenges pose a threat to the sustainability and quality of care and underscore the need to explore innovative solutions.

An alternative model of care that has been identified as a possible solution is the Hospital at Home (HaH) model, in which healthcare professionals actively deliver hospital-level care in patients' homes for a limited period of time[8]. There are two main types of HaH programmes— early supported discharge (ESD) and admission avoidance (AA). The former aims to accelerate the discharge of admitted patients who still require hospital-level services (partial substitution of inpatient care), whereas the latter recruits patients directly from emergency departments or GP referrals (complete substitution).

One of the initial impetuses behind explorations of the HaH model was to reduce adverse events associated with the hospitalisation of older patients[9]. Individual studies have demonstrated the benefits of HaH programmes in terms of safety[9 10], effectiveness[11 12], reduced iatrogenic complications[13] and cost savings[14 15]. Several systematic reviews have also been conducted, but evidence for some outcomes remains unclear or conflicting, partly due to the differences in approach between reviews. Greater clarity may be attained by consolidating the various reviews[16].

In addition, despite past suggestions that outcomes such as costs and clinical complications vary between ESD and AA models due to differing extents of inpatient care substitution[17-19], these have not been comprehensively assessed. A 2017 NICE Guideline[17] presented analyses according to programme type, but the further stratification of results based on level of care precluded any clear conclusions.

While a review of reviews has been conducted for alternative strategies to inpatient hospitalisation[20], there has not been a review specifically for HaH. The objectives of the current

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review were to synthesise the evidence for safety and effectiveness of HaH according to programme type, and assess whether one of the two types of HaH programme may potentially be more beneficial for health systems that are grappling with capacity constraints and increasing costs. This would guide prioritisation in service planning and focus future research efforts. Process indicators and facilitators and barriers that would be of interest to potential adopter organisations were also summarised.

2. METHODS

2.1 Data Sources and Searches

A search was performed on 14 January 2020 using PubMed, EMBASE, Cochrane Database of Systematic Reviews, Web of Science and Scopus. The search was restricted to systematic reviews published after January 2005. While HaH pilot studies have been implemented before that, the care model took time to mature and was largely inconsistently defined prior to a seminal paper in 2005[9]. Hence, reviews after 2005 were considered more representative of the HaH literature. The search strategy employed terms that operationalised "review" and were related to "Hospital at Home" (Table 1). A pre-specified update of the search was performed on 8 June 2020. There was no registered protocol for this study.

Table 1: Search syntax for database search

Database	Number	Search
PubMed	#1	Search: ((((((home care services, hospital-based [mesh]) OR ("hospital at home")) OR ("home hospital*")) OR ("early supported discharge" [Title/Abstract])) OR ("admission avoidance" [Title/Abstract])) OR (((home[Title/Abstract]) AND hospital[Title/Abstract]))) AND ((((review[Title/Abstract] OR overview[Title/Abstract])) OR meta-analy*[Title/Abstract]) OR meta-review*[Title/Abstract]) Filters: English, from 2005 – 2020
Scopus	#1	((TITLE-ABS-KEY (hospital* W/2 home)) OR (ALL ("home hospital")) OR (ALL ("hospital at home")) OR (TITLE-ABS-KEY (early W/1 support* W/1 discharge)) OR (TITLE-ABS-KEY (admission W/1 avoid*))) AND ((TITLE-ABS-KEY (overview OR review) OR TITLE-ABS-KEY (meta AND analy*) OR TITLE-ABS-KEY ("meta review") OR TITLE-ABS-KEY ("review of reviews"))) AND PUBYEAR > 2004 AND (LIMIT-TO (SUBJAREA , "MEDI") OR LIMIT-TO (SUBJAREA , "NURS") OR LIMIT-TO (SUBJAREA , "SOCI") OR LIMIT-TO (SUBJAREA , "HEAL")) AND (LIMIT-TO (LANGUAGE , "English")
Cochrane	#1	MeSH descriptor: [Home Care Services, Hospital-Based] explode all trees
	#2	MeSH descriptor: [Home Care Services] explode all trees
	#3	MeSH descriptor: [Hospitalization] explode all trees
	#4	#2 and #3
	#5	(early NEAR/1 support* NEAR/1 discharge):ab,ti
	#6	(admission NEAR/1 avoid*):ab,ti
	#7	(hospital* NEAR/3 home):ab,ti,kw

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	#8	"hospital at home"
	#9	"home hospital*"
	#10	#1 or #4 or #5 or #6 or #7 or #8 or #9 with Cochrane Library publication date from Jan 2005 to Feb 2020, in Cochrane Reviews
Web of Science	#1	ts= (hospital* NEAR/3 home)
	#2	ts = "early support* discharge"
	#3	ts = "admission avoid*"
	#4	ts = "home hospital*"
	#5	Ts = "hospital at home"
	#6	#5 OR #4 OR #3 OR #2 OR #1
	#7	ts= (overview OR review OR "meta analy*" OR "meta review" OR "review of reviews")
	#8	#7 AND #6
	#9	ESCI Timespan=2005-2020
EMBASE	#1	'home care'/exp
	#2	'hospital'/exp
	#3	#1 AND #2
	#4	(hospital* NEAR/3 home):ab,ti,kw
	#5	'early support* discharge':ab,ti,kw
	#6	'admission avoid*':ab,ti,kw
	#7	#3 OR #4 OR #5 OR #6
	#8	overview:ab,ti OR review:ab,ti OR 'meta analy*':ab,ti OR 'meta review':ab,ti OR 'review of reviews':ab,ti
	#9	#7 AND #8
	#10	#7 AND #8 AND [english]/lim AND [2005-2020]/py

2.2 Study Selection

We included systematic reviews of HaH interventions where healthcare professionals provided hospital-level care in patients' homes for acute and post-acute conditions, and which reported at least one of the following: clinical outcomes, resource use or process indicators. Systematic reviews including other interventions were eligible if subgroup analyses for HaH were presented. English-language systematic reviews of all study designs were eligible.

We excluded reviews of palliative, obstetric, paediatric and mental health HaH schemes as separate reviews are justified for these groups[21]. We excluded reviews that were entirely overlapped by a subsequent update or review of higher quality.

2.3 Data Extraction and Quality Assessment

Review characteristics and relevant outcomes were extracted from the reported findings and tables in the reviews. Primary outcomes of interest were mortality, readmissions, costs, and length of stay (LOS). Secondary outcomes of interest were patient and caregiver satisfaction and caregiver

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burden. Process indicators (recruitment rates, adverse events) and facilitators and barriers to implementation were also extracted.

Methodological quality of the reviews was assessed using the Assessment of Multiple Systematic Reviews- 2 (AMSTAR-2) checklist[22] which provides overall ratings (high, moderate, low, critically low) based on weaknesses in critical domains. Methodological quality of the primary studies was determined using the review authors' original assessments of risk of bias.

2.4 Data Synthesis and Analysis

Findings from included reviews were grouped, narratively summarised and compared. Due to heterogeneity between studies, a meta-analysis was not possible.

2.5 Patient Involvement

Patients or the public were not involved in the conceptualisation or carrying out of this research. Ethics review was not sought and patient consent was not required for this publication.

3. RESULTS

The search returned 7869 potentially relevant references. No relevant papers were identified from the search update. Ten reviews were eventually included. The PRISMA flow diagram in Figure 1 details the article selection process.

3.1 Review Characteristics

The ten reviews (four Cochrane reviews, six non-Cochrane reviews) were published between 2012 and 2017. Cumulatively, there were 100 relevant primary studies, with substantial overlap in studies between reviews (40%). The studies were published between 1976 and 2016, and were mostly from the UK, US, Australia, Italy and Spain.

Some reviews assessed either ESD or AA interventions, while several reviews included both ESD and AA studies. The reviews were categorised according to three use cases: ESD[21 23], AA[8 24 25] and ESD/AA[26-30]. Characteristics of the studies are summarized in Table 2.

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Table 2: Characteristics of included reviews

S/N	Review	Date assessed as up-to-date	Study objectives	Participant characteristics	No. of relevant studies[a]	Study design (No. of studies)	Risk of bias of primary studies	AMSTAR-2 rating
Early Supported Discharge								
1	Goncalves-Bradley et al,[21] 2017	Jan 2017	To determine the effectiveness and cost of HaH compared with inpatient hospital care.	Elderly patients with a mix of medical conditions (e.g. cardio-respiratory diseases, neurological conditions, fractures), stroke patients, and elective surgical patients (e.g. hip and knee replacements)	32	RCT (32)	Majority of the trials were at low risk of bias.	High
2	Langhorne et al,[23] 2017	Apr 2017	To assess the effect of ESD on hospital LOS, patient and carer outcomes and resource use in comparison with conventional hospital care.	Stroke patients	15	RCT (15)	Trials were generally at low risk of bias, but blinding was impossible.	High
Admission Avoidance								
4	Shepperd et al,[8] 2016	Mar 2016	To determine the effectiveness and cost of managing patients with admission avoidance HaH compared with inpatient hospital care.	Patients with COPD, stroke, cellulitis, community-acquired pneumonia, fever, neutropenia, dementia, neuromuscular disease and other acute medical conditions	16	RCT (16)	Majority of the studies were at low risk of bias, but unclear risk of bias for selective reporting and performance bias.	High
5	Qaddoura et al,[24] 2015	Oct 2014	To assess the effect of HaH on clinical, patient-centred, and cost outcomes in heart failure in comparison to hospital care.	Patients with heart failure	5	RCT (3), before-after study (2)	RCTs were modest in quality. Blinding was not possible for all studies. Several studies had unclear risk of selection bias, high risk of attrition bias, and lack of protocol standardisation. Lack of control for confounders in observational studies.	Moderate

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6	Varney et al,[25] 2014	2013	To assess the efficacy of AA HaH services that admit patients directly from the ED in comparison to hospital care.	Patients with COPD, cellulitis, pneumonia, heart failure, dementia with acute illness, deep vein thrombosis and other conditions.	16	RCT (9), case-control study (1), observational (5), cost-analysis (1)	Majority of the studies were at low risk of bias.	Low
Early Supported Discharge and Admission Avoidance								
7	Caplan et al,[26] 2012	Feb 2012	To assess the effect of HaH services that significantly substitute for in-hospital time (≥ 7 days or ≥ 25% of LOS of inpatient controls) on mortality, readmissions, patient and carer satisfaction, and costs.	Medical, surgical and rehabilitation patients >16 years (conditions not specified)	56	RCT (56)	Almost all studies were not blinded.	Low
8	Sriskandarajah et al,[30] 2018	Sep 2016	To review observational studies evaluating the safety and effectiveness of HaH and OPAT.	Patients with a mix of conditions (conditions not specified)	6	Observational (6)	Risk of bias ranged from low to high; majority of studies had moderate risk of bias.	Low
7	Echevarria et al,[27] 2016	Nov 2014	To assess the safety, efficacy and cost of ESD and AA compared to hospital care for patients with acute exacerbation of COPD	Patients with acute exacerbation of COPD	8	RCT (8)	Low risk of selection bias; high risk of performance bias on subjective outcomes as blinding was not possible; risk of attrition bias in 6 trials.	Moderate
8	Jeppesen et al,[28] 2012	Oct 2010	To evaluate the efficacy of HaH compared to hospital inpatient care in acute exacerbations of COPD.	Patients with acute exacerbation of COPD	8	RCT (8)	Majority of trials had low risk of bias; high risk of bias in 1 trial due to selective reporting and "other biases" (not described).	High
9	McCurdy[29] 2012	Aug 2010	To compare HaH with inpatient hospital care for patients with acute exacerbations of COPD who present to the emergency department (ED).	Patients with acute exacerbation of COPD	6	RCT (6)	Majority of the studies had high risk of bias due to lack of allocation concealment, unclear methods for randomisation, unclear blinding of assessors, inadequate sample sizes and improper ITT analyses (withdrawals/dropouts ignored).	Moderate

[a]Primary studies that did not fulfil eligibility criteria were excluded from this count.

Abbreviations: LOS = Length of stay. COPD = chronic obstructive pulmonary disease. OPAT = outpatient parenteral antimicrobial therapy.

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3.2 Quality of Studies

There were four reviews[8 21 23 28] of high quality, five reviews[24-27 29] of moderate quality, and one review[30] of low quality. Main areas of weakness were the lack of a registered protocol and failure to provide a list of excluded studies (refer to Supplement Table 1 for AMSTAR-2 item ratings).

The randomised trials were mostly assessed to be at low risk of bias, while the observational studies were of moderate to high risk of bias.

3.3 Outcomes

Detailed findings and summary measures are reported in Table 3.

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Table 3: Detailed primary outcomes in ESD, AA and ESD/AA

Review	Mortality	Readmissions	LOS	Costs and cost-effectiveness
ESD				
Goncalves-Bradley et al,[21] 2017	Stroke patients: RR 0.92 at 3 to 6 months (95% CI 0.57 to 1.48, P = 0.71; 11 trials; moderate-quality evidence). Mix of conditions: RR 1.07 at 3 to 6 months (95% CI 0.76 to 1.49, P = 0.71; 8 trials; moderate-quality evidence). COPD patients: RR 0.53 (95% CI 0.25 to 1.12, P = 0.096; 5 trials; low-quality evidence). Elective surgery patients: little or no difference (range of -3.8% to 0.1%; 3 trials; low-quality evidence).	Stroke patients: RR 1.09 (95% CI 0.71 to 1.66, P = 0.70; 5 trials; low-quality evidence). Mix of conditions: RR 1.25 at 3 months (95% CI 0.98 to 1.58, P = 0.069; 9 trials; moderate-quality evidence). COPD patients: RR 0.86 (95% CI 0.66 to 1.13, P = 0.29; 5 trials; low-quality evidence). Elective surgery patients: little or no difference (5 trials; low-quality evidence)	*Hospital LOS* Stroke patients: MD -6.68 days (95% CI -10.19 to -3.17, P < 0.001; 4 trials; moderate-quality evidence). Median reductions of 8 to 15 days in 6 other trials. Mix of conditions: MD -0.36 to -22 days (8 trials). Pooled data from 4 trials: MD -6.76 days (95% CI -10.60 to -2.92, P < 0.001; moderate-quality evidence). Orthopaedic surgery patients: MD -4.44 days (95% CI -6.37 to -2.51, P < 0.00001; 4 trials; moderate-quality evidence). *Total LOS* Mix of conditions: MD 6.43 days (95% CI 2.84 to 10.03, P < 0.001; 3 trials). Elective surgery patients: MD 2.79 days (95% CI 0.77 to 4.81; 2 trials).	Initial and total costs were similar, lower or higher than inpatient care across trials (low to very low-quality evidence). Only one trial factored in community costs; mean hospital cost savings (at 6 months post-randomisation) were reduced from AUD 4678 to AUD 2013 upon factoring in community costs, primarily due to the costs of home-based rehabilitation (MD: AUD 2906)
Langhorne et al,[23] 2017	OR 1.04 at 6 months (95% CI 0.77 to 1.40, P = 0.81; 16 studies; moderate-quality evidence). Higher mortality in the subgroup without co-ordinated ESD teams than the subgroup with co-ordinated ESD teams (P = 0.01). Reduced odds	OR 1.09 at 6 months (95% CI 0.79 to 1.51, P = 0.59; 6 trials; low-quality evidence).	MD -5.5 days (95% CI -2.9 to -8.2, P < 0.0001; 16 trials; moderate-quality evidence). Larger reduction for severe stroke subgroup (MD -28 days, 95% CI -17 to -40) than moderate severity subgroup (MD -3 days, 95% CI -1 to -7), P < 0.0001.	Total costs ranged from 23% less to 15% greater than inpatient care (7 trials).

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	of death or dependency in the moderate stroke subgroup (OR 0.77, 95% CI 0.61 to 0.98) than the severe stroke subgroup (OR1.40, 95% CI 0.83 to 2.36); P = 0.04.			
AA				
Shepperd et al,[8] 2016	RR 0.77 at 3 months (95% CI 0.54 to 1.09, P = 0.15; 5 trials; moderate-quality evidence). RR 0.77 at 6 months (95% CI 0.60 to 0.99, P = 0.04; 6 trials; moderate-quality evidence).	RR 0.98 at 3 to 12 months (95% CI 0.77 to 1.23, P = 0.84; 7 trials; moderate-quality evidence).	MD -8.09 to 15.90 days (7 trials)	Trend towards lower initial and total costs (8 trials). Estimated MD in health service costs varied, from GBP -210.90 at 3 months to GBP -304.72 per episode. In 1 trial, cost differences were no longer significant upon including the costs of informal care, although HaH might still be cheaper (MD: GBP -2216, 95% CI GBP -4771 to GBP 339).
Qaddoura et al,[24] 2015	RR 0.94 (95% CI 0.67 to 1.32); P = 0.18; 3 trials). Significant reduction in 1 observational study (P < 0.05).	RR 0.68 (95% CI 0.42 to 1.09, P = 0.34; 2 trials); non-significant reduction in 1 trial; significant reductions in 2 observational studies (MD -3.80 to -0.70 readmissions per patient). Similar total readmission rates (including readmissions during HaH care) to inpatient care in 1 study.	Significantly longer LOS in 3 studies (MD 3.0 to 9.1 days)	Significant reductions in costs per episode (mean/median reduction €295.97 to €2691; 3 trials); trend towards lower total costs at 12 months (mean/median reduction €3070 to €3125; 2 trials).
Varney et al,[25] 2014	No significant differences across 5 trials.	Low rates of unplanned hospital admissions across 4 observational studies.	Longer LOS across 3 trials, with significant effects in 2 trials.	Lower costs across 6 studies, with significant effects in 4 studies; statistical significance not reported in the other 2 studies.
ESD/AA				

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Caplan et al,[26] 2012	Overall: OR 0.81 (95% CI 0.69 to 0.95, P =0.008; 42 studies) Medical patients: OR 0.79 (95% CI 0.65 to 0.97, P = 0.02; 23 studies). Surgical patients: OR 0.78 (95% CI 0.29 to 2.10, P = 0.62; 5 studies). Rehabilitation patients: OR 0.83 (95% CI 0.63 to 1.08, P = 0.17; 13 studies).	Overall: OR 0.75 (95% CI 0.59 to 0.95, P = 0.02; 41 studies; χ^2 = 73.27, P = 0.001). Medical patients: OR 0.76 (95% CI 0.60 to 0.97, P = 0.02; 18 studies). Surgical patients: OR 0.66 (95% C, 0.36 to 1.22, P = 0.18; 10 studies). Rehabilitation patients: OR 0.96 (95% CI 0.70 to 1.31, P = 0.79; 9 studies).	NR	Proportional differences in costs revealed lower costs under HaH. Overall: MD -1567.11 (95% CI -2069.53 to -1064.69, P < 0.001; χ^2 = 237.45, P < 0.001). Medical patients: MD -1821.69 (95% CI -2591.89 to -1051.49, P < 0.001; 5 studies). Surgical patients: MD -875.89, 95% (CI -3510.96 to 1759.19, P = 0.21; 2 studies). Rehabilitation patients: MD -2704.18 (95% CI -3973.29 to -1435.07, P < 0.001; 4 studies).
Sriskandarajah et al,[30] 2018	NR	Readmission rates during HaH care ranged from 4.2% to 9.7% (5 studies).	NR	NR
Echevarria et al,[27] 2016	RR 0.66 at 2 to 6 months (95% CI 0.40 to 1.09, P = 0.10; 7 trials).	Post-HaH discharge readmissions at 2 to 6 months: RR 0.74 at 2 to 6 months (95% CI 0.60 to 0.90, P = 0.003; 7 trials). Total readmissions: RR 0.84 (95% CI 0.69 to 1.01, P = 0.07; 7 trials).	NR	Lower costs per episode (4 trials), but 1 trial reported higher costs when societal costs (over 3 months) were factored in (HaH: €6,304; Control: €5,395), and a savings per QALY lost of €31,111.
Jeppesen et al,[28] 2012	RR 0.65 at 2 to 6 months (95% CI 0.40 to 1.04, P = 0.07; 7 trials; moderate-quality evidence).	RR 0.76 at 1 to 6 months, 95% CI 0.59 to 0.99, P = 0.04; 8 trials; moderate-quality evidence).	NR	Trend towards reductions in direct costs (3 trials; very low-quality evidence).
McCurdy[29] 2012	RR 0.68 at 2 to 6 months (95% CI 0.41 to 1.12, P = 0.13; 6 trials).	Total readmissions: RR 0.90 (95% CI 0.70 to 1.16; P = 0.41; 6 trials). Readmissions during HaH period accounted for 13% to 50% of total readmissions.	Total LOS was similar to inpatient care in 2 trials (MD 0.15 to 1.5 days), shorter in 1 trial (MD -2.9 days), and longer in 2 trials (mean/median difference 2 to 4.5 days).	NR

All risk ratios (RR), odds ratios (OR) and mean differences (MD) are for HaH compared with inpatient care.
Abbreviations: LOS = Length of stay. NR = Not reported.

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3.3.1 ESD

Clinical Outcomes

Two reviews[21 23] reported that ESD did not significantly affect the mortality rates of elderly patients with a mix of conditions (moderate-quality evidence), stroke patients (moderate-quality evidence), and elective surgical patients (low-quality evidence).

One review[21] revealed a trend towards higher readmissions in ESD schemes for a mix of conditions (moderate-quality evidence), whereas readmissions did not differ significantly between ESD and inpatient groups for stroke and surgical patients (low-quality evidence). It was unclear if these analyses included readmissions during ESD care.

Resource Use and Costs

Two reviews[21 23] concluded that ESD significantly reduced hospital LOS for all patient groups (moderate-quality evidence), although high heterogeneity in results was observed. Goncalves et al[21] observed a significant increase in total LOS (inclusive of the stay in hospital and HaH) across three trials for a mix of conditions.

Two reviews[21 23] found conflicting findings for the effect of ESD on costs (very low quality evidence). High heterogeneity in cost estimates was also observed due to differences in intervention, cost components, and cost measurements across trials. Only one trial factored in community costs; mean hospital cost savings under ESD (at six months post-randomisation) were reduced from AUD 4678 to AUD 2013 upon factoring in those costs.

Patient and Caregiver Outcomes

Two reviews[21 23] reported that ESD generally led to similar or improved levels of patient and caregiver satisfaction and caregiver strain when compared to inpatient care (low-quality evidence). However, for surgical patients, three out of five trials observed lower caregiver satisfaction under ESD[21].

Process indicators

Recruitment rates were low for all patient groups, with a median of 20% to 33% of patients meeting eligibility criteria[21 23]. Regarding adverse events and clinical complications, one review[21]

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reported a lack of difference between ESD and inpatient groups in three trials for surgical patients, but significantly lower rates of delirium in the ESD group in a trial for a mix of conditions (P = 0.003).

3.3.2 AA

Clinical Outcomes

AA may reduce mortality rates compared to inpatient care. A Cochrane review[8] for a mix of conditions reported a trend towards lower mortality under AA care at three months, and a significant reduction in mortality at six months (moderate-quality evidence). Two other reviews[24 25] did not find significant effects on mortality, but both reviews raised the issue of under-powering.

A review for heart failure[24] observed a trend towards fewer readmissions (post-discharge from HaH) under HaH programmes. One of the studies reported total readmissions (inclusive of readmissions during HaH care), which was comparable between AA and inpatient groups. Another review[8] pooled results for total readmissions, and did not find any significant difference between groups (moderate-quality evidence).

Resource Use and Costs

Two reviews[24 25] reported longer LOS under HaH schemes. Another review[8] concluded that the effect on LOS was unclear due to high heterogeneity (mean difference: -8.09 to 17 days), but there was a general trend towards longer LOS.

Three reviews[8 24 25] found that HaH reduced costs for the acute episode, with this effect reaching significance in several studies (low-quality evidence). Estimated mean cost savings per episode was GBP 304.72 in one review[8], and ranged from €295.97 to €2691 in another review[24]. However, Shepperd et al[8] suggested that the cost benefit may be offset by the costs of informal care as one trial revealed that the cost reduction was no longer significant upon including informal care costs. Two reviews[8 24] revealed a trend towards lower total costs (inclusive of follow-up costs) under HaH, although Qaddoura et al[24] concluded that the effect was unclear as few studies included indirect costs.

Patient and Caregiver Outcomes

One review[8] observed significantly higher levels of patient satisfaction in HaH than inpatient care across most trials (low-quality evidence). Two other reviews[24 25] also found high levels of satisfaction among HaH patients in observational studies. For caregiver outcomes, Shepperd et al[8]

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concluded that there was insufficient evidence to draw conclusions as two trials revealed conflicting results.

Process Indicators

One review[8] reported that recruitment rates for HaH were low but no data was provided. One review[25] concluded that HaH generally did not affect the rate of adverse events. Shepperd et al[8] reported a lack of difference in clinical complications in one trial, and significantly lower rates of bowel and urinary complications and behavioural problems (in dementia patients) under HaH in two trials.

3.3.3 ESD/AA

Clinical Outcomes

One review[26] for a mix of conditions reported significantly lower mortality under HaH as compared to inpatient care. However, this analysis should be interpreted with caution as several of the included primary studies did not meet the criteria of HaH (e.g. nature of care was long-term care or active care was not provided by healthcare professionals[31-34]). Subgroup analysis by the degree of inpatient care substitution did not reveal notable differences. Three reviews for COPD patients[27-29] reported a trend towards lower mortality under HaH (moderate-quality evidence).

Caplan et al[26] reported that HaH significantly reduced readmissions compared to inpatient care. However, interpretation of this result is constrained as the analysis included four irrelevant studies (psychiatric studies), and this subgroup had the largest effect size amongst the various specialties. It is also unclear if readmissions during HaH care were included in these analyses. For COPD patients, two reviews[27 28] found significant reductions in post-HaH discharge readmissions (moderate-quality evidence)[27], while two reviews[27 29] observed non-significant reductions in total readmissions[27].

Healthcare Resources and Costs

One review[29] concluded that the effect of HaH on total LOS for COPD patients could not be determined due to conflicting findings.

Caplan et al[26] reported that HaH significantly reduced costs compared to inpatient care, albeit with significant heterogeneity. The cost components of these analyses were unclear. Two reviews for

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COPD patients[27 28] concluded that evidence of cost savings under HaH was of very low quality due to heterogeneity. Echevarria et al[27] reported that one study observed higher costs for HaH than inpatient care when societal costs were included. The study also found that HaH was associated with a savings per QALY lost of €31,111.

Patient and Caregiver Outcomes

One review[26] reported that majority of the trials revealed higher patient and caregiver satisfaction, and lower caregiver burden under HaH. Two reviews[28 29] reported that the effect of HaH on patient satisfaction for COPD patients was inconclusive due to limited data (low quality evidence). One trial observed that HaH did not affect caregiver satisfaction[28].

Process Indicators

COPD-specific reviews[27-29] reported low recruitment rates, with the percentage of eligible patients ranging from 11% to 39% across trials. 56% to 88.5% of eligible patients consented to participation[29].

McCurdy et al[29] concluded that HaH did not affect the rate of clinical complications, although there were lower rates of urinary tract infections under HaH in one trial. A review of observational studies[30] observed higher rates of complications in the HaH group than the inpatient group in one study, but the HaH group was also seven times larger. In another study, patients receiving IV antibiotics under AA were at lower risk of adverse effects than ESD patients.

3.4 Facilitators and Barriers

Facilitators to implementation include coordinated and multi-disciplinary HaH teams[8 23]. Barriers to implementation include home proximity to the hospital[28 29], medical condition stability[8 28 29], and level of disability[21 23]. ESD led to greater reductions in death or dependency[23] and higher cost-effectiveness in one trial[21] when limited to patients with mild or moderate disability. In addition, caregiver availability was an eligibility criteria in one-third to half of the studies across reviews[8 21].

4. DISCUSSION

To our knowledge, this is the first systematic review of reviews for HaH. While HaH is commonly defined to be the provision of acute care at home, the current paper defined HaH as hospital-level care for both acute and post-acute conditions. Services for the latter mostly comprise

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interventions which can also be separately classified as "rehabilitation-at-home", but we included these under HaH, in accordance with the approach by Cochrane reviews which have also included trials providing both acute and post-acute services[35 36], or post-acute care alone[37-39].

There were some notable findings for each type of HaH programme. ESD generally resulted in mortality and readmission rates similar to inpatient care, but there was an unexpected trend towards higher readmissions in schemes for a mix of conditions. ESD significantly reduced hospital LOS, but with a trend towards increased total LOS– which is commonly due to reduced time constraints and more time to deliver personalised care within the home[40]. Findings for costs were conflicting.

AA programmes resulted in a trend towards lower mortality rates which reached significance at six months, and readmission rates that were either comparable or non-significantly lower compared to inpatient care— depending on the inclusion of transfers back to the hospital during HaH care. While there was a general trend towards increased LOS, initial and total costs of AA schemes were lower than inpatient treatment in absolute or statistically significant terms.

Among the ESD/AA reviews, a meta-analysis[26] revealed significant benefits for mortality, readmission and costs under HaH. This review could be insightful in providing a broad overview of HaH; however, as cautioned above, several of the included trials did not meet the criteria of HaH. Three other reviews for COPD[27-29] reported a trend towards lower mortality and readmissions under HaH, while evidence for LOS and costs was insufficient or unclear.

4.1 Comparison of ESD and AA

Between the two HaH programme types, ESD has been better studied, but AA programmes potentially lead to greater benefits in terms of costs, and possibly clinical outcomes as well. Although the evidence on costs was generally low across reviews, the consistent cost benefits observed across AA studies are promising— in contrast to the inconclusive findings of ESD reviews. These observations are in line with propositions that AA may result in greater cost savings than ESD due to the complete substitution of inpatient stay[25].

Regarding clinical outcomes, AA interventions also appear more promising when considering the trend towards lower mortality, coupled with the concerning trend of higher readmissions in ESD trials for a mix of conditions. The former could have been driven by reductions in iatrogenic events

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such as delirium and infections[25 26]— this is one of the commonly suggested benefits of AA over ESD. However, apart from a study in one review[30] suggesting this, insufficient evidence regarding adverse events precluded clear conclusions.

Nonetheless, with the existing evidence, and given the complete substitution of inpatient stay in AA interventions, we would recommend the prioritisation of AA programmes in further explorations of the HaH model. This is pertinent to health systems that are grappling with shortages of hospital beds and rising healthcare spending. However, we acknowledge that these indirect and post-hoc comparisons are limited, especially given the heterogeneity in implementation and patient groups across studies. Hence, it should be cautioned that the strength of evidence for our overarching conclusion is low.

4.2 Recommendations for Practice

Regardless of programme type, when compared to inpatient care, HaH is generally associated with similar or improved clinical outcomes, shorter hospital LOS, and high levels of patient satisfaction. HaH has been widely implemented within several countries including the UK and Australia, with government reports from these countries encouraging the implementation and expansion of HaH services to relieve the bed crunch in hospitals and to manage rising costs[17 41].

A limited number of facilitators and barriers to implementation have been summarised here, but additional factors brought up in studies which were not included also deserve mention. A provider evaluation of HaH[42] highlighted that many countries where HaH has been widely implemented have single-payer systems and strong imperatives to keep medical costs low, whereas greater barriers are evident in countries such as the US where payment norms are still predominantly fee-for-service. Nonetheless, with the shift towards value-based payments[43] and integrated care[44] in various health systems, a more conducive environment for the HaH model may be emerging.

Also, HaH is more practical in urban areas with a sufficient and predictable patient load to allow for greater economies of scale [17 45 46]. A 2011 Australian economic evaluation estimated that a 10% increase in HaH enrolment from public hospitals would result in government savings of $1.7 million[47]. Given the low recruitment rates in HaH schemes, one of the included reviews[29] recommended scaling up HaH programmes by expanding hospital boundaries in urban areas and

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recruiting patients during evenings and weekends, as several trials only recruited patients during particular hours of the day and/or weekdays.

Lastly, at the programme level, a recent primary study[40] reported initial resistance against HaH from physicians over safety and liability concerns. Reflecting this, other studies have highlighted the importance of obtaining buy-in through active clinician engagement, well-defined responsibilities, clear communication lines and cross-team collaboration[48 49]. Another potential facilitator is telemedicine. A recent trial[50] provided virtual consultations and 24-hour access to video, and suggested that this allowed the handling of high acuity cases and a high degree of medical input.

4.3 Evidence gaps

The most well-studied diagnostic groups were COPD and stroke, with clear evidence of benefits to clinical outcomes for COPD patients. Past research has indicated that other conditions are also particularly suitable for HaH treatment, such as cellulitis and deep-vein thrombosis[51 52]. However, there was a lack of subgroup analyses or reviews for these conditions, hence the effects of HaH interventions for these groups could be further clarified.

It was unclear in multiple reviews if early readmissions during HaH care were factored into analyses for readmissions, hence it was challenging to make comparisons across reviews. Future reviews should clearly report both early and total readmissions. The evidence on costs was of low quality due to heterogeneity across studies, and few studies assessed cost-effectiveness. Future studies could consider full economic evaluations that include all direct and indirect costs (e.g. informal care costs) to better guide policy and decision making. Further clarification on caregiver outcomes, especially caregiver burden, is also required given caregivers' crucial roles in HaH programmes[53].

Lastly, there was limited evidence on process indicators such as adverse events. Given that the reduction of iatrogenic events is an often-cited point of comparison between ESD and AA models, and also one of the primary motivations of developing HaH models, future primary studies and reviews should seek to report these.

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4.4 Limitations

Outcomes were mainly limited by small sample sizes and heterogenous measurement tools in the primary studies. Also, two reviews[23 26] included primary studies that did not meet the criteria of HaH, hence there were likely inaccuracies in the reported meta-analyses.

The current research might have missed primary studies that were not included in the systematic reviews, and we also excluded grey literature. However, we attempted to address these limitations by highlighting relevant findings from primary studies and government publications.

Also, overlaps in primary studies across reviews could have inflated results. This limitation was most pronounced for COPD-specific reviews (90% overlap), but these reviews were included as they provided insight on different outcomes.

5. Conclusion

HaH is largely safe and effective for suitable patients requiring hospital-level care, and clearly warrants greater attention in health systems that are challenged by capacity constraints and rising costs. However, several outcomes and process indicators still require further clarification, such as caregiver outcomes, cost-effectiveness and clinical complications. Based on preliminary comparisons, AA schemes should be prioritised in explorations of the HaH model due to the complete substitution of hospital stay and possible cost savings which will likely increase with scaled-up programmes.

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Figure 1. PRISMA flow diagram for article selection.

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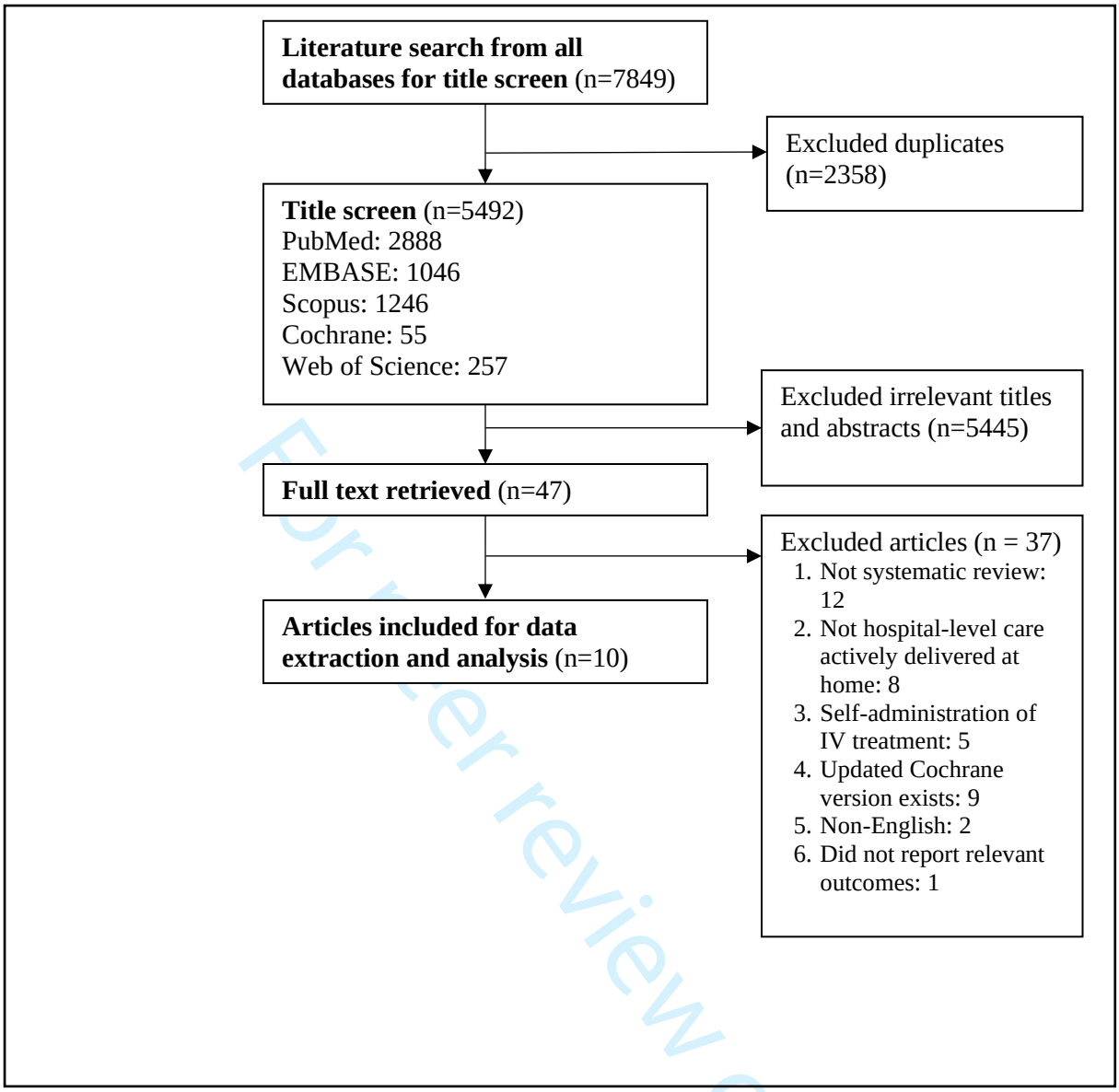


Figure 1: PRISMA flow diagram for article selection

22211

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Supplementary Table 1: AMSTAR-2 ratings

Review	1. PICO	2. A-priori*	3. Inclusion rationale for study design	4. Search strategy[a]	5. Study selection in duplicate	6. Data extraction in duplicate	7. List of excluded studies and justification	8. Detailed description of included study	9. Assessed RoB in included studies[a]
Goncalves-Bradley et al,[21] 2017	Y	Y	N	Y	Y	Y	Y	Y	Y
Langhorne et al,[9] 2017	Y	Y	N	Y	Y	Y	Y	Y	Y
Shepperd et al,[8] 2016	Y	Y	N	Y	Y	Y	Y	Y	Y
Qaddoura et al,[25] 2015	Y	P	N	Y	Y	Y	N	Y	Y
Varney et al,[26] 2014	Y	P	Y	P	N	N	N	Y	P
Caplan et al,[27] 2012	Y	P	N	P	Y	Y	N	N	Y
Sriskandarajah et al,[31] 2018	Y	P	Y	P	N	N	N	N	P
Echevarria et al,[28] 2016	Y	P	N	P	Y	N	N	Y	Y
Jeppesen et al,[29] 2012	Y	Y	N	Y	Y	Y	Y	Y	Y
McCurdy[30] 2012	Y	P	N	P	N	N	N	Y	Y

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Hospital at Home systematic review of reviews

Review	10. Sources of funding of included studies	11. Appropriate methods for meta-analyses	12. Reviewed impact of RoB on meta-analysis	13. Accounted for RoB when discussing results[a]	14. Explained heterogeneity	15. Investigated publication bias in meta-analyses	16. Reported conflict of interest	Overall Rating[b]
Goncalves-Bradley et al,[21] 2017	Y	Y	Y	Y	Y	Y	Y	High
Langhorne et al,[9] 2017	N	Y	Y	Y	Y	Y	Y	High
Shepperd et al,[8] 2016	N	Y	Y	Y	Y	Y	Y	High
Qaddoura et al,[25] 2015	Y	Y	N	Y	Y	Y	Y	Moderate
Varney et al,[26] 2014	N	NA	N	Y	Y	NA	Y	Moderate
Caplan et al,[27] 2012	N	Y	N	Y	Y	Y	N	Moderate
Sriskandarajah et al,[31] 2018	N	NA	NA	Y	N	NA	Y	Low
Echevarria et al,[28] 2016	N	Y	Y	Y	Y	N	Y	Moderate
Jeppesen et al,[29] 2012	N	Y	Y	Y	Y	N	Y	High
McCurdy[30] 2012	N	Y	N	Y	Y	N	Y	Moderate

[a]Critical item.

Abbreviations: Y = Yes, P = Partial, N = No, RoB = Risk of Bias.

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PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Nil
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Nil
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8-9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	11-13
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Nil
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Nil
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Nil
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	18
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	20-21
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	18-19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

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Hospital at Home: A Systematic Review of Reviews

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HOSPITAL AT HOME: A SYSTEMATIC REVIEW OF REVIEWS

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ABSTRACT

Objectives: To provide an overview of the safety and effectiveness of Hospital-at-Home (HaH) according to HaH programme type (early-supported discharge (ESD) versus admission avoidance (AA)), and identify the model which has greater potential to benefit health systems grappling with capacity constraints and rising costs.

Methods: A systematic review of reviews was conducted by performing a search on PubMed, EMBASE, Cochrane Database of Systematic Reviews, Web of Science and Scopus (January 2005 to June 2020) for English-language systematic reviews evaluating HaH. Data on primary outcomes (mortality, readmissions, costs, length of stay (LOS)), secondary outcomes (patient/caregiver outcomes), and process indicators were extracted. Quality of the reviews was assessed using AMSTAR-2.

Results: Ten systematic reviews were identified, with four of high quality, five of moderate quality, and one of low quality. The reviews were classified according to three use cases. ESD reviews generally revealed comparable mortality and readmission rates to inpatient care, shorter hospital LOS, increased total LOS (including stay at home), and unclear findings for costs. AA reviews observed a trend towards lower mortality and costs, comparable or lower readmissions, and a trend towards longer LOS (at home under HaH). Among reviews including both programme types (ESD/AA), chronic obstructive pulmonary disease reviews revealed improved clinical outcomes but unclear findings for resource use.

Conclusion: For suitable patients, HaH largely results in similar or improved clinical outcomes compared to inpatient treatment, and can potentially mitigate some of the challenges facing health systems. Preliminary comparisons suggest that AA models could be prioritised over ESD in further explorations of HaH due to potential benefits in costs and clinical outcomes. Nonetheless, future research should clarify costs of HaH programmes given the current low-quality evidence, and also address evidence gaps pertaining to caregiver outcomes and adverse events under HaH care.

Protocol registration: There was no registered protocol.

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Strengths and limitations of this study

- The first systematic review of reviews providing a comprehensive overview of the safety and efficacy of HaH
- Highlights evidence gaps pertaining to outcomes and process indicators in the current HaH literature
- Compares two major HaH models to offer relevant recommendations for health systems facing capacity constraints and rising costs
- Strength of evidence for comparison between HaH models is low due to heterogeneity in implementation and patient groups across studies

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1. INTRODUCTION

Healthcare systems around the world are grappling with ageing populations, increasing healthcare needs and expenditures, and shortages of hospital beds and manpower[1 2]. Such capacity constraints decrease the efficiency and quality of care delivery, and can lead to delays in elective and emergency treatment[3], increased mortality risk[4 5], lower patient and staff satisfaction, and higher costs for both patients and the health service[6]. Chronic morbidity in ageing populations have further driven the rise in healthcare costs[7]. These complex challenges pose a threat to the sustainability and quality of care and underscore the need to explore innovative solutions.

An alternative model of care that has been identified as a possible solution is the Hospital at Home (HaH) model, in which healthcare professionals deliver hospital-level care in patients' homes for a limited period of time[8]. There are two main types of HaH programmes— early supported discharge (ESD) and admission avoidance (AA). The former aims to accelerate the discharge of admitted patients who still require hospital-level services (partial substitution of hospital care), whereas the latter directly admits patients into HaH based on GP referrals thereby avoiding physical contact with the hospital, or through direct admission from the emergency room without inpatient stay. One of the initial impetuses behind the HaH model was to reduce adverse events associated with the hospitalisation of older patients[9]. Individual studies have demonstrated the benefits of HaH programmes in terms of safety[9 10], effectiveness[11 12], reduced iatrogenic complications[13] and cost savings[14 15]. Several systematic reviews have also been conducted, but evidence for some outcomes remains unclear or conflicting, partly due to different approaches between reviews. Greater clarity may be attained by consolidating the reviews[16].

In addition, despite past suggestions that outcomes such as costs and clinical complications vary between ESD and AA models due to differing extents of inpatient care substitution[17-19], these have not been comprehensively assessed. A 2017 NICE Guideline[17] presented analyses according to programme type, but the further stratification of results based on level of care precluded any clear conclusions.

While a review of reviews has been conducted for alternative strategies to inpatient hospitalisation[20], there has not been a review specifically for HaH. The objectives of this review were

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to synthesise the evidence for safety and effectiveness of HaH according to programme type, and assess whether one of the two types of HaH programme may potentially be more beneficial for health systems that are grappling with capacity constraints and increasing costs. This would guide prioritisation in service planning and focus future research efforts. Process indicators and facilitators and barriers to implementation were also summarised. These are pertinent for potential adopter organisations to evaluate the feasibility of HaH in their respective settings as past studies have pointed to constraints in the HaH model (e.g. in terms of patient eligibility due to medical condition severity[21] and caregiver availability[22]). Moreover, process indicators are valuable in assessing the fidelity of implementation and identifying components that increase the success of HaH implementation[23].

2. METHODS

2.1 Data Sources and Searches

A search was performed on 14 January 2020 using PubMed, EMBASE, Cochrane Database of Systematic Reviews, Web of Science and Scopus. Search terms and criteria for inclusion/exclusion were determined a priori by all authors. Test searches were carried out and results were reviewed and refined by two authors before finalising the search strategy. The search was restricted to systematic reviews published after January 2005. While HaH pilot studies have been implemented before that, the care model took time to mature and was largely inconsistently defined prior to a seminal paper in 2005[9]. Hence, reviews after 2005 were considered more representative of the HaH literature. The search strategy employed terms that operationalised "review" and were related to "Hospital at Home" (Table 1). A pre-specified update of the search was performed on 8 June 2020. There was no registered protocol for this study.

Table 1: Search syntax for database search

Database	Number	Search
PubMed	#1	Search: ((((((home care services, hospital-based [mesh]) OR ("hospital at home")) OR ("home hospital*")) OR ("early supported discharge" [Title/Abstract])) OR ("admission avoidance" [Title/Abstract])) OR (((home[Title/Abstract]) AND hospital[Title/Abstract]))) AND ((((review[Title/Abstract] OR overview[Title/Abstract])) OR meta-analy*[Title/Abstract]) OR meta-review*[Title/Abstract]) Filters: English, from 2005 – 2020
Scopus	#1	((TITLE-ABS-KEY (hospital* W/2 home)) OR (ALL ("home hospital")) OR (ALL ("hospital at home")) OR (TITLE-ABS-KEY (early W/1 support* W/1 discharge)) OR (TITLE-ABS-KEY (admission W/1 avoid*))) AND ((TITLE-ABS-KEY (overview OR review) OR TITLE-ABS-KEY (meta

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		AND analy*) OR TITLE-ABS-KEY ("meta review") OR TITLE-ABS-KEY ("review of reviews"))) AND PUBYEAR > 2004 AND (LIMIT-TO (SUBJAREA , "MEDI") OR LIMIT-TO (SUBJAREA , "NURS") OR LIMIT-TO (SUBJAREA , "SOCI") OR LIMIT-TO (SUBJAREA , "HEAL")) AND (LIMIT-TO (LANGUAGE , "English")
Cochrane	#1	MeSH descriptor: [Home Care Services, Hospital-Based] explode all trees
	#2	MeSH descriptor: [Home Care Services] explode all trees
	#3	MeSH descriptor: [Hospitalization] explode all trees
	#4	#2 and #3
	#5	(early NEAR/1 support* NEAR/1 discharge):ab,ti
	#6	(admission NEAR/1 avoid*):ab,ti
	#7	(hospital* NEAR/3 home):ab,ti,kw
	#8	"hospital at home"
	#9	"home hospital*"
	#10	#1 or #4 or #5 or #6 or #7 or #8 or #9 with Cochrane Library publication date from Jan 2005 to Feb 2020, in Cochrane Reviews
Web of Science	#1	ts= (hospital* NEAR/3 home)
	#2	ts = "early support* discharge"
	#3	ts = "admission avoid*"
	#4	ts = "home hospital*"
	#5	Ts = "hospital at home"
	#6	#5 OR #4 OR #3 OR #2 OR #1
	#7	ts= (overview OR review OR "meta analy*" OR "meta review" OR "review of reviews")
	#8	#7 AND #6
	#9	ESCI Timespan=2005-2020
EMBASE	#1	'home care'/exp
	#2	'hospital'/exp
	#3	#1 AND #2
	#4	(hospital* NEAR/3 home):ab,ti,kw
	#5	'early support* discharge':ab,ti,kw
	#6	'admission avoid*':ab,ti,kw
	#7	#3 OR #4 OR #5 OR #6
	#8	overview:ab,ti OR review:ab,ti OR 'meta analy*':ab,ti OR 'meta review':ab,ti OR 'review of reviews':ab,ti
	#9	#7 AND #8
	#10	#7 AND #8 AND [english]/lim AND [2005-2020]/py

2.2 Study Selection

We included systematic reviews of HaH interventions, with HaH defined as the active provision of care by healthcare professionals in patients' homes for acute and post-acute conditions that otherwise would require hospital inpatient care. Care services should only be provided for a time-limited period[8 24]. Reviews of interventions that were delivered in outpatient settings, provided long-term care, or primarily involved patient self-care at home were excluded. Systematic reviews including other interventions were eligible if subgroup analyses for HaH were presented. We included reviews which reported at least one of the following: clinical outcomes, resource use or process

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indicators. Process indicators were defined as measures of the processes involved in achieving programme objectives[25]. English-language systematic reviews of all study designs were eligible.

We excluded reviews of palliative, obstetric, paediatric and mental health HaH schemes due to differences in these groups' care needs and objectives from general medicine patients. We excluded reviews that were entirely overlapped by a subsequent update or review of higher quality, as determined using the Assessment of Multiple Systematic Reviews- 2 (AMSTAR-2)[26]

Literature search and shortlist of the articles were done consistently by one investigator, and shortlisted articles were reviewed by two authors for final inclusion.

2.3 Data Extraction and Quality Assessment

Review characteristics and relevant outcomes were extracted from the reported findings and tables in the reviews. Primary outcomes of interest were mortality, readmissions, costs, and length of stay (LOS). As the entire treatment duration under HaH is considered inpatient stay, LOS in ESD schemes includes both the stay in the hospital and home. For AA schemes, LOS refers to LOS in the home before discharge from HaH. Secondary outcomes of interest were patient and caregiver satisfaction and caregiver burden. Process indicators (recruitment rates, adverse events) and facilitators and barriers to implementation were also extracted.

Methodological quality of the reviews was assessed using AMSTAR-2[26] which provides overall ratings (high, moderate, low, critically low) based on weaknesses in critical domains. In line with Cochrane guidelines[27], methodological quality of the primary studies was determined using the review authors' original assessments of risk of bias.

2.4 Data Synthesis and Analysis

Findings from included reviews were grouped, narratively summarised and compared. Due to heterogeneity between studies, a meta-analysis was not possible.

2.5 Patient Involvement

Patients or the public were not involved in this research. Ethics review was not sought and patient consent was not required.

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3. RESULTS

The search returned 7869 potentially relevant references. No relevant papers were identified from the search update. Ten reviews were eventually included. The PRISMA flow diagram in Figure 1 details the article selection process.

3.1 Review Characteristics

The ten reviews (four Cochrane reviews, six non-Cochrane reviews) were published between 2012 and 2017. Cumulatively, there were 100 relevant primary studies, with substantial overlap in studies between reviews (40%). The studies were published between 1976 and 2016, and were mostly from the UK, US, Australia, Italy and Spain.

Some reviews either assessed only ESD or AA interventions, while several reviews included both ESD and AA studies. The reviews were categorised according to three use cases: ESD[24 28], AA[8 29 30] and ESD/AA[31-35]. Characteristics of the reviews are summarized in Table 2. Service structure of the HaH programmes in the primary studies varied both within and between reviews. There was heterogeneity in the team coordinating care (e.g. hospital versus community-based), care team composition (e.g. nurses, physiotherapists, occupational therapists), programme components (e.g. additional elements such as patient and caregiver education), number of home visits, availability of after-hours support, and specific medical services provided (e.g. home oxygen, IV fluids).

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Table 2: Characteristics of included reviews

S/N	Review	Date assessed as up-to-date	Study objectives	Participant characteristics	No. of relevant studies[a]	Study design (No. of studies)	Risk of bias of primary studies	AMSTAR-2 rating
Early Supported Discharge								
1	Goncalves-Bradley et al,[24] 2017	Jan 2017	To determine the effectiveness and cost of HaH compared with inpatient hospital care.	Elderly patients with a mix of medical conditions (e.g. cardio-respiratory diseases, neurological conditions, fractures), stroke patients, and elective surgical patients (e.g. hip and knee replacements)	32	RCT (32)	Majority of the trials were at low risk of bias.	High
2	Langhorne et al,[28] 2017	Apr 2017	To assess the effect of ESD on hospital LOS, patient and carer outcomes and resource use in comparison with conventional hospital care.	Stroke patients	15	RCT (15)	Trials were generally at low risk of bias, but blinding was impossible.	High
Admission Avoidance								
4	Shepperd et al,[8] 2016	Mar 2016	To determine the effectiveness and cost of managing patients with admission avoidance HaH compared with inpatient hospital care.	Patients with COPD, stroke, cellulitis, community-acquired pneumonia, fever, neutropenia, dementia, neuromuscular disease and other acute medical conditions	16	RCT (16)	Majority of the studies were at low risk of bias, but unclear risk of bias for selective reporting and performance bias.	High
5	Qaddoura et al,[29] 2015	Oct 2014	To assess the effect of HaH on clinical, patient-centred, and cost outcomes in heart failure in comparison to hospital care.	Patients with heart failure	5	RCT (3), before-after study (2)	RCTs were modest in quality. Blinding was not possible for all studies. Several studies had unclear risk of selection bias, high risk of attrition bias, and lack of protocol standardisation. Lack of control for confounders in observational studies.	Moderate

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6	Varney et al,[30] 2014	2013	To assess the efficacy of AA HaH services that admit patients directly from the ED in comparison to hospital care.	Patients with COPD, cellulitis, pneumonia, heart failure, dementia with acute illness, deep vein thrombosis and other conditions.	16	RCT (9), case-control study (1), observational (5), cost-analysis (1)	Majority of the studies were at low risk of bias.	Low
Early Supported Discharge and Admission Avoidance								
7	Caplan et al,[31] 2012	Feb 2012	To assess the effect of HaH services that significantly substitute for in-hospital time (≥ 7 days or ≥ 25% of LOS of inpatient controls) on mortality, readmissions, patient and carer satisfaction, and costs.	Medical, surgical and rehabilitation patients >16 years (conditions not specified)	56	RCT (56)	Almost all studies were not blinded.	Low
8	Sriskandarajah et al,[35] 2018	Sep 2016	To review observational studies evaluating the safety and effectiveness of HaH and OPAT.	Patients with a mix of conditions (conditions not specified)	6	Observational (6)	Risk of bias ranged from low to high; majority of studies had moderate risk of bias.	Low
7	Echevarria et al,[32] 2016	Nov 2014	To assess the safety, efficacy and cost of ESD and AA compared to hospital care for patients with acute exacerbation of COPD	Patients with acute exacerbation of COPD	8	RCT (8)	Low risk of selection bias; high risk of performance bias on subjective outcomes as blinding was not possible; risk of attrition bias in 6 trials.	Moderate
8	Jeppesen et al,[33] 2012	Oct 2010	To evaluate the efficacy of HaH compared to hospital inpatient care in acute exacerbations of COPD.	Patients with acute exacerbation of COPD	8	RCT (8)	Majority of trials had low risk of bias; high risk of bias in 1 trial due to selective reporting and "other biases" (not described).	High
9	McCurdy[34] 2012	Aug 2010	To compare HaH with inpatient hospital care for patients with acute exacerbations of COPD who present to the emergency department (ED).	Patients with acute exacerbation of COPD	6	RCT (6)	Majority of the studies had high risk of bias due to lack of allocation concealment, unclear methods for randomisation, unclear blinding of assessors, inadequate sample sizes and improper ITT analyses (withdrawals/dropouts ignored).	Moderate

[a] Primary studies that did not fulfil eligibility criteria were excluded from this count.

Abbreviations: LOS = Length of stay. COPD = chronic obstructive pulmonary disease. OPAT = outpatient parenteral antimicrobial therapy.

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3.2 Quality of Studies

There were four reviews[8 24 28 33] of high quality, five[29-32 34] of moderate quality, and one [35] of low quality. Main areas of weakness were the lack of a registered protocol and failure to provide a list of excluded studies (refer to Supplement Table 1 for AMSTAR-2 ratings).

The randomised trials were mostly assessed to be at low risk of bias, while the observational studies were of moderate to high risk of bias.

3.3 Outcomes

Detailed findings and summary measures are reported in Table 3.

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Table 3: Detailed primary outcomes in ESD, AA and ESD/AA

Review	Mortality	Readmissions	LOS	Costs and cost-effectiveness
ESD				
Goncalves-Bradley et al,[24] 2017	Stroke patients: RR 0.92 at 3 to 6 months (95% CI 0.57 to 1.48, P = 0.71; 11 trials; moderate-quality evidence). Mix of conditions: RR 1.07 at 3 to 6 months (95% CI 0.76 to 1.49, P = 0.71; 8 trials; moderate-quality evidence). COPD patients: RR 0.53 (95% CI 0.25 to 1.12, P = 0.096; 5 trials; low-quality evidence). Elective surgery patients: little or no difference (range of -3.8% to 0.1%; 3 trials; low-quality evidence).	Stroke patients: RR 1.09 (95% CI 0.71 to 1.66, P = 0.70; 5 trials; low-quality evidence). Mix of conditions: RR 1.25 at 3 months (95% CI 0.98 to 1.58, P = 0.069; 9 trials; moderate-quality evidence). COPD patients: RR 0.86 (95% CI 0.66 to 1.13, P = 0.29; 5 trials; low-quality evidence). Elective surgery patients: little or no difference (5 trials; low-quality evidence)	*Hospital LOS* Stroke patients: MD -6.68 days (95% CI -10.19 to -3.17, P < 0.001; 4 trials; moderate-quality evidence). Median reductions of 8 to 15 days in 6 other trials. Mix of conditions: MD -0.36 to -22 days (8 trials). Pooled data from 4 trials: MD -6.76 days (95% CI -10.60 to -2.92, P < 0.001; moderate-quality evidence). Orthopaedic surgery patients: MD -4.44 days (95% CI -6.37 to -2.51, P < 0.00001; 4 trials; moderate-quality evidence).*Total LOS (with home LOS)* Mix of conditions: MD 6.43 days (95% CI 2.84 to 10.03, P < 0.001; 3 trials). Elective surgery patients: MD 2.79 days (95% CI 0.77 to 4.81; 2 trials).	Initial and total costs were similar, lower or higher than inpatient care across trials (low to very low-quality evidence). Only one trial factored in community costs; mean hospital cost savings (at 6 months post-randomisation) were reduced from AUD 4678 to AUD 2013 upon factoring in community costs, primarily due to the costs of home-based rehabilitation (MD: AUD 2906)
Langhorne et al,[28] 2017	OR 1.04 at 6 months (95% CI 0.77 to 1.40, P = 0.81; 16 studies; moderate-quality evidence). Reduced odds of death or dependency in the moderate stroke subgroup (OR 0.77, 95% CI 0.61 to 0.98) than the severe stroke subgroup (OR	OR 1.09 at 6 months (95% CI 0.79 to 1.51, P = 0.59; 6 trials; low-quality evidence).	MD -5.5 days (95% CI -2.9 to -8.2, P < 0.0001; 16 trials; moderate-quality evidence). Larger reduction for severe stroke subgroup (MD -28 days, 95% CI -17 to -40) than moderate severity subgroup (MD -3 days, 95% CI -1 to -7), P < 0.0001.	Total costs ranged from 23% less to 15% greater than inpatient care (7 trials).

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	1.40, 95% CI 0.83 to 2.36); P = 0.04.			
AA				
Shepperd et al,[8] 2016	RR 0.77 at 3 months (95% CI 0.54 to 1.09, P = 0.15; 5 trials; moderate-quality evidence). RR 0.77 at 6 months (95% CI 0.60 to 0.99, P = 0.04; 6 trials; moderate-quality evidence).	RR 0.98 at 3 to 12 months (95% CI 0.77 to 1.23, P = 0.84; 7 trials; moderate-quality evidence).	MD -8.09 to 15.90 days (7 trials)	Trend towards lower initial and total costs (8 trials). Estimated MD in health service costs varied, from GBP -210.90 at 3 months to GBP -304.72 per episode. In 1 trial, cost differences were no longer significant upon including the costs of informal care, although HaH might still be cheaper (MD: GBP -2216, 95% CI GBP -4771 to GBP 339).
Qaddoura et al,[29] 2015	RR 0.94 (95% CI 0.67 to 1.32); P = 0.18; 3 trials). Significant reduction in 1 observational study (P < 0.05).	RR 0.68 (95% CI 0.42 to 1.09, P = 0.34; 2 trials); non-significant reduction in 1 trial; significant reductions in 2 observational studies (MD -3.80 to -0.70 readmissions per patient). Similar total readmission rates (including readmissions during HaH care) to inpatient care in 1 study.	Significantly longer LOS in 3 studies (MD 3.0 to 9.1 days)	Significant reductions in costs per episode (mean/median reduction €295.97 to €2691; 3 trials); trend towards lower total costs at 12 months (mean/median reduction €3070 to €3125; 2 trials).
Varney et al,[30] 2014	No significant differences across 5 trials.	Low rates of unplanned hospital admissions across 4 observational studies.	Longer LOS across 3 trials, with significant effects in 2 trials.	Lower costs across 6 studies, with significant effects in 4 studies; statistical significance not reported in the other 2 studies.
ESD/AA				
Caplan et al,[31] 2012	Overall: OR 0.81 (95% CI 0.69 to 0.95, P =0.008; 42 studies) Medical patients: OR 0.79 (95% CI 0.65 to 0.97, P = 0.02; 23 studies).	Overall: OR 0.75 (95% CI 0.59 to 0.95, P = 0.02; 41 studies; χ^2 = 73.27, P = 0.001). Medical patients: OR 0.76 (95% CI 0.60 to 0.97, P = 0.02; 18 studies). Surgical patients: OR 0.66 (95% C, 0.36 to 1.22, P = 0.18; 10 studies).	NR	Proportional differences in costs revealed lower costs under HaH. Overall: MD -1567.11 (95% CI -2069.53 to -1064.69, P < 0.001; χ^2 = 237.45, P < 0.001). Medical patients: MD -1821.69 (95% CI -2591.89 to -1051.49, P < 0.001; 5 studies).

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	Surgical patients: OR 0.78 (95% CI 0.29 to 2.10, P = 0.62; 5 studies). Rehabilitation patients: OR 0.83 (95% CI 0.63 to 1.08, P = 0.17; 13 studies).	Rehabilitation patients: OR 0.96 (95% CI 0.70 to 1.31, P = 0.79; 9 studies).		Surgical patients: MD -875.89, 95% (CI -3510.96 to 1759.19, P = 0.21; 2 studies). Rehabilitation patients: MD -2704.18 (95% CI -3973.29 to -1435.07, P < 0.001; 4 studies).
Sriskandarajah et al,[35] 2018	NR	Readmission rates during HaH care ranged from 4.2% to 9.7% (5 studies).	NR	NR
Echevarria et al,[32] 2016	RR 0.66 at 2 to 6 months (95% CI 0.40 to 1.09, P = 0.10; 7 trials).	Post-HaH discharge readmissions at 2 to 6 months: RR 0.74 at 2 to 6 months (95% CI 0.60 to 0.90, P = 0.003; 7 trials). Total readmissions: RR 0.84 (95% CI 0.69 to 1.01, P = 0.07; 7 trials).	NR	Lower costs per episode (4 trials), but 1 trial reported higher costs when societal costs (over 3 months) were factored in (HaH: €6,304; Control: €5,395), and a savings per QALY lost of €31,111.
Jeppesen et al,[33] 2012	RR 0.65 at 2 to 6 months (95% CI 0.40 to 1.04, P = 0.07; 7 trials; moderate-quality evidence).	RR 0.76 at 1 to 6 months, 95% CI 0.59 to 0.99, P = 0.04; 8 trials; moderate-quality evidence).	NR	Trend towards reductions in direct costs (3 trials; very low-quality evidence).
McCurdy[34] 2012	Overall: RR 0.68 at 2 to 6 months (95% CI 0.41 to 1.12, P = 0.13; 6 trials). ESD: RR 0.33 (95% CI 0.13 to 0.85, P = 0.02; 3 trials). AA: RR 0.85 (95% CI 0.45 to 1.62, P = 0.63; 2 studies)	Total readmissions: RR 0.90 (95% CI 0.70 to 1.16; P = 0.41; 6 trials). Readmissions during HaH period accounted for 13% to 50% of total readmissions.	Total LOS was similar to inpatient care in 2 trials (MD 0.15 to 1.5 days), shorter in 1 trial (MD -2.9 days), and longer in 2 trials (mean/median difference 2 to 4.5 days).	NR

All risk ratios (RR), odds ratios (OR) and mean differences (MD) are for HaH compared with inpatient care.
Abbreviations: LOS = Length of stay. NR = Not reported.

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3.3.1 ESD

Clinical Outcomes

Two reviews[24 28] reported that ESD did not significantly affect the mortality rates of elderly patients with a mix of conditions (moderate-quality evidence), stroke patients (moderate-quality evidence), and elective surgical patients (low-quality evidence).

One review[24] revealed a trend towards higher readmissions in ESD schemes for a mix of conditions (moderate-quality evidence), whereas readmissions did not differ significantly between ESD and inpatient groups for stroke and surgical patients (low-quality evidence). It was unclear if these analyses included readmissions during ESD care.

Resource Use and Costs

Two reviews[24 28] concluded that ESD significantly reduced hospital LOS for all patient groups (moderate-quality evidence), although high heterogeneity in results was observed. Goncalves et al[24] observed a significant increase in total LOS (in the hospital and home) across three trials for a mix of conditions.

Two reviews[24 28] found conflicting findings for the effect of ESD on costs (very low-quality evidence). High heterogeneity in cost estimates was also observed due to differences in intervention, cost components, and cost measurements across trials. Only one trial factored in community costs; mean hospital cost savings under ESD (at six months post-randomisation) were reduced from AUD 4678 to AUD 2013 upon factoring in those costs.

Patient and Caregiver Outcomes

Two reviews[24 28] reported that ESD generally led to similar or improved levels of patient and caregiver satisfaction and caregiver strain when compared to inpatient care (low-quality evidence). However, for surgical patients, three out of five trials observed lower caregiver satisfaction under ESD[24].

Process indicators

Recruitment rates were low for all patient groups, with a median of 20% to 33% of patients meeting eligibility criteria[24 28]. Regarding adverse events and clinical complications, one review[24]

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reported a lack of difference between ESD and inpatient groups in three trials for surgical patients, but significantly lower rates of delirium in the ESD group in a trial for a mix of conditions (P = 0.003).

3.3.2 AA

Clinical Outcomes

AA may reduce mortality rates compared to inpatient care. A Cochrane review[8] for a mix of conditions reported a trend towards lower mortality under AA care at three months, and a significant reduction in mortality at six months (moderate-quality evidence). Two other reviews[29 30] did not find significant effects on mortality, but both reviews raised the issue of under-powering.

A review for heart failure[29] observed a trend towards fewer readmissions (post-discharge from HaH) under HaH programmes. One of the studies reported total readmissions (inclusive of readmissions during HaH care), which was comparable between AA and inpatient groups. Another review[8] pooled results for total readmissions, and did not find any significant difference between groups (moderate-quality evidence).

Resource Use and Costs

Two reviews[29 30] reported longer LOS (at home) under HaH schemes. Another review[8] concluded that the effect on LOS was unclear due to high heterogeneity (mean difference: -8.09 to 17 days), but there was a general trend towards longer LOS.

Three reviews[8 29 30] found that HaH reduced costs for the acute episode, with this effect reaching significance in several studies (low-quality evidence). Estimated mean cost savings per episode was GBP 304.72 in one review[8], and ranged from €295.97 to €2691 in another review[29]. However, Shepperd et al[8] suggested that the cost benefit may be offset by the costs of informal care as one trial revealed that the cost reduction was no longer significant upon including informal care costs. Two reviews[8 29] revealed a trend towards lower total costs (inclusive of follow-up costs) under HaH, although Qaddoura et al[29] concluded that the effect was unclear as few studies included indirect costs.

Patient and Caregiver Outcomes

One review[8] observed significantly higher levels of patient satisfaction in HaH than inpatient care across most trials (low-quality evidence). Two other reviews[29 30] also found high levels of satisfaction among HaH patients in observational studies. For caregiver outcomes, Shepperd et al[8]

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concluded that there was insufficient evidence to draw conclusions as two trials revealed conflicting results.

Process Indicators

One review[8] reported that recruitment rates for HaH were low but no data was provided. One review[30] concluded that HaH generally did not affect the rate of adverse events. Shepperd et al[8] reported a lack of difference in clinical complications in one trial, and significantly lower rates of bowel and urinary complications and behavioural problems (in dementia patients) under HaH in two trials.

3.3.3 ESD/AA

Clinical Outcomes

One review[31] for a mix of conditions reported significantly lower mortality under HaH as compared to inpatient care. However, this analysis should be interpreted with caution as several of the included primary studies did not meet the criteria of HaH (e.g. nature of care was long-term care or active care was not provided by healthcare professionals[36-39]). Three reviews for COPD patients[32-34] reported a trend towards lower mortality under HaH (moderate-quality evidence). Subgroup analysis in one review[34] revealed that the non-significant reduction was maintained for AA, but reached significance in ESD trials.

Caplan et al[31] reported that HaH significantly reduced readmissions compared to inpatient care. However, interpretation of this result is constrained as the analysis included four irrelevant studies (psychiatric studies), and this subgroup had the largest effect size amongst the various specialties. It is also unclear if readmissions during HaH care were included in these analyses. For COPD patients, two reviews[32 33] found significant reductions in post-HaH discharge readmissions (moderate-quality evidence)[32], while two reviews[32 34] observed non-significant reductions in total readmissions[32]. Subgroup analysis[34] by programme type did not reveal differences in readmission rates.

Healthcare Resources and Costs

One review[34] concluded that the effect of HaH on total LOS for COPD patients could not be determined due to conflicting findings.

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Caplan et al[31] reported that HaH significantly reduced costs compared to inpatient care, albeit with significant heterogeneity. The cost components of these analyses were unclear. Two reviews for COPD patients[32 33] concluded that evidence of cost savings under HaH was of very low quality due to heterogeneity. Echevarria et al[32] reported that one study observed higher costs for HaH than inpatient care when societal costs were included. The study also found that HaH was associated with a savings per QALY lost of €31,111.

Patient and Caregiver Outcomes

One review[31] reported that majority of the trials revealed higher patient and caregiver satisfaction, and lower caregiver burden under HaH. Two reviews[33 34] reported that the effect of HaH on patient satisfaction for COPD patients was inconclusive due to limited data (low quality evidence). One trial observed that HaH did not affect caregiver satisfaction[33].

Process Indicators

COPD-specific reviews[32-34] reported low recruitment rates, with the percentage of eligible patients ranging from 11% to 39% across trials. 56% to 88.5% of eligible patients consented to participation[34].

McCurdy et al[34] concluded that HaH did not affect the rate of clinical complications, although there were lower rates of urinary tract infections under HaH in one trial. A review of observational studies[35] observed higher rates of complications in the HaH group than the inpatient group in one study, but the HaH group was also seven times larger. In another study, patients receiving IV antibiotics under AA were at lower risk of adverse effects than ESD patients.

3.4 Facilitators and Barriers

Facilitators to implementation include coordinated and multi-disciplinary HaH teams[8 28]. Barriers to implementation include physical distance of patients' homes to the hospital[33 34], medical condition stability[8 33 34], and level of disability[24 28]. ESD led to greater reductions in death or dependency[28] and higher cost-effectiveness in one trial[24] when limited to patients with mild or moderate disability. In addition, caregiver availability was an eligibility criteria in one-third to half of the studies across reviews[8 24].

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4. DISCUSSION

To our knowledge, this is the first systematic review of reviews for HaH. While HaH is commonly defined as the provision of acute care at home, the current paper defined HaH as hospital-level care for both acute and post-acute conditions. Services for the latter mostly comprise interventions which can also be classified as "rehabilitation-at-home", but we included these under HaH, in accordance with Cochrane reviews which have also included trials providing both acute and post-acute services[40 41], or post-acute care alone[42-44].

In comparing ESD and AA reviews, it is evident that ESD has been better studied, but AA potentially leads to greater benefits in terms of clinical outcomes and costs. AA programmes resulted in a trend towards lower mortality rates which reached significance at six months, and readmission rates that were either comparable or non-significantly lower compared to inpatient care— depending on the inclusion of transfers back to the hospital during HaH care. On the other hand, mortality and readmission rates in ESD programmes were generally similar to inpatient care, but with a concerning trend of higher readmissions in schemes for a mix of conditions. The observed benefit of AA over ESD with regards to mortality and readmissions could have been driven by reductions in iatrogenic events such as delirium and infections[30 31]— this is a commonly suggested benefit of AA over ESD. However, apart from a study in one review[35] suggesting less iatrogenic events under AA compared to ESD, insufficient evidence regarding adverse events precluded clear conclusions.

In addition, AA reduced the initial and total costs of care in absolute or statistically significant terms when compared to usual treatment, whereas the evidence for costs was conflicting for ESD. Although the quality of evidence was generally low across reviews, the consistent cost benefits observed across AA studies are promising when contrasted with the inconclusive findings in ESD studies. These observations are aligned with propositions that AA may result in greater cost savings than ESD due to the complete substitution of hospital stay[30].

Regarding LOS, although AA completely substituted hospital stay and ESD significantly reduced hospital LOS, it is noted that there was a trend towards increased total LOS for both AA and ESD when compared to usual care. The increased total LOS could have been due to the inclusion of post-acute care in some HaH schemes[45 46]; in traditional hospitalisations, patients would likely have

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been transferred to step-down care facilities such as community hospitals and skilled nursing facilities which are not factored into LOS calculations. It is also worth highlighting that some implementations[47] realised cost reductions despite longer LOS, possibly due to lower fixed costs from traditional hospitalisations.

Among the ESD/AA reviews, one meta-analysis[31] found significant benefits for mortality, readmissions and costs under HaH, with subgroup analyses by degree of hospital stay substitution (more than or less than 50%) revealing little difference in mortality rates. However, as cautioned above, this meta-analysis is challenging to interpret as several of the included trials did not meet the operational definition of HaH. Also, while the subgroup analysis could have been insightful in our comparison of ESD and AA, the 50% criterion employed did not adequately capture the distinction between the two. COPD-specific ESD/AA reviews[32-34] reported a trend towards lower mortality and readmissions under HaH. One review[34] conducted subgroup analyses on clinical outcomes by programme type, but the small number of studies (n = 5) precluded conclusive interpretation. ESD/AA reviews with planned subgroup analyses for programme type should continue to be explored in future research.

Overall, considering all the reviews, HaH is generally associated with similar or improved clinical outcomes, shorter hospital LOS, and high levels of patient satisfaction compared to inpatient care. It is recommended to prioritise AA implementations in further explorations of the HaH model due to the potential benefits of AA over ESD in terms of clinical outcomes and costs. Moreover, given the complete substitution of hospital stay under AA, this recommendation is pertinent to health systems that are grappling with bed shortages and rising healthcare spending. Another practical benefit of AA over ESD is the reduced need for repeated patient data transfer between hospital units and lower risk of relevant information being lost when the patient is directly admitted to the home.

Nonetheless, it should be cautioned that the strength of evidence for our recommendation is low due to the limitations of our indirect and post-hoc comparisons, especially given the heterogeneity in implementation and patient groups across studies. We also acknowledge that it may not always be possible to maintain a clear distinction between ESD and AA in real-life implementations due to operational considerations.

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4.2 Recommendations for Practice

At a broad level, contextual factors for consideration regarding HaH implementation include hospital location and payment structures. Given the low recruitment rates in HaH schemes, HaH may be more practical in urban areas with a sufficient and predictable patient load to allow for greater economies of scale [17 48 49]. Secondly, many countries where HaH has been widely implemented (e.g. Australia, UK) have single-payer systems and strong imperatives to keep medical costs low, whereas greater barriers are evident in countries such as the US where payment norms are still predominantly episodic and fee-for-service[50]. Nonetheless, with the shift towards value-based payments[51] and integrated care[52] across health systems, a more conducive environment for HaH may be emerging.

At the programme level, care enablers include the proximity of patients' homes to the hospital and suitability of home settings. With the greater impetus to decentralise care and bring care closer to patients' homes during the COVID-19 pandemic, it may be timely to re-examine how future urban planning and housing designs can support models of care such as HaH. Care enablers identified in primary studies which were not included in the current paper deserve mention as well. These include skills-training of staff to provide care in HaH schemes[53 54] and assistive technologies such as telemedicine. Regarding the latter, a recent trial[55] provided virtual consultations and 24-hour access to video, and suggested that this allowed the handling of high acuity cases and a high degree of medical input.

Several reviews highlighted that HaH may not be suitable for patients with intensive care needs or who do not live with caregivers. The latter appears to present a dilemma that warrants further investigation. While necessitating caregiver availability may reduce the risks associated with patients living alone (e.g. delayed medical assistance), a possible concern is the transfer of burden of care from hospitals to family caregivers. However, by incorporating appropriate safeguards (e.g. panic buttons, wearable technologies such as fall-detection sensors), caregiver reliance can in fact be reduced while still ensuring that patients receive quality care. Hence, moving beyond the traditional model of reliance on care providers, HaH can potentially serve as an avenue for patient empowerment by increasing patient involvement in their own care, boosting their confidence in self-management, and

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improving care outcomes. Such programme design would also extend HaH eligibility to patients without caregivers, thus allowing more to benefit from HaH schemes.

4.3 Evidence gaps

The most well-studied diagnostic groups were COPD and stroke, with clear evidence of benefits to clinical outcomes for COPD patients. Past research has indicated that other conditions are also particularly suitable for HaH treatment, such as cellulitis and deep-vein thrombosis[56 57]. However, there was a lack of subgroup analyses or reviews for these conditions, hence the effects of HaH interventions for these groups could be further clarified.

It was unclear in multiple reviews if early readmissions during HaH care were factored into analyses for readmissions, hence it was challenging to make comparisons across reviews. Future reviews should clearly report both early and total readmissions. The evidence on costs was of low quality due to heterogeneity across studies, and few studies assessed cost-effectiveness. Future studies could consider full economic evaluations that include all direct and indirect costs (e.g. informal care costs) to better guide policy and decision making. Further clarification on caregiver outcomes, especially caregiver burden, is also required given caregivers' crucial roles in HaH programmes[22].

Lastly, there was limited reporting on facilitators, barriers and process indicators such as adverse events. Given that the reduction of iatrogenic events is an often-cited point of comparison between ESD and AA models, and also one of the primary motivations of developing HaH models, future primary studies and reviews should seek to report these.

4.4 Limitations

The first limitation is the lack of a registered protocol for this study. In addition, the current research might have missed primary studies that were not included in the systematic reviews, and we also excluded grey literature. However, we attempted to address these limitations by highlighting relevant findings from primary studies and government publications.

Outcomes were mainly limited by small sample sizes and heterogenous measurement tools in the primary studies. Also, two reviews[28 31] included primary studies that did not meet the criteria of HaH, hence there were likely inaccuracies in the reported meta-analyses.

For peer review only

Lastly, overlaps in primary studies across reviews could have inflated results. This limitation was most pronounced for COPD-specific reviews (90% overlap), but these reviews were included as they provided insight on different outcomes.

5. Conclusion

HaH is largely safe and effective for suitable patients requiring hospital-level care. HaH generally leads to similar or improved clinical outcomes, shorter hospital LOS, and high patient satisfaction when compared to inpatient care, and clearly warrants greater attention in health systems that are challenged by capacity constraints and rising costs. With the ongoing COVID-19 pandemic, there is in fact greater urgency to explore models of care such as HaH that allow bed capacities to ramp up quickly by decentralising and shifting care back to the communities. Moreover, based on preliminary comparisons, AA schemes should be prioritised in further explorations of HaH due to the complete substitution of hospital stay, and possible cost savings which will likely increase with scaled-up programmes. However, for both AA and ESD, several outcomes and process indicators still require further clarification, such as caregiver outcomes, cost-effectiveness and clinical complications.

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Figure 1. PRISMA flow diagram for article selection.

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For peer review only

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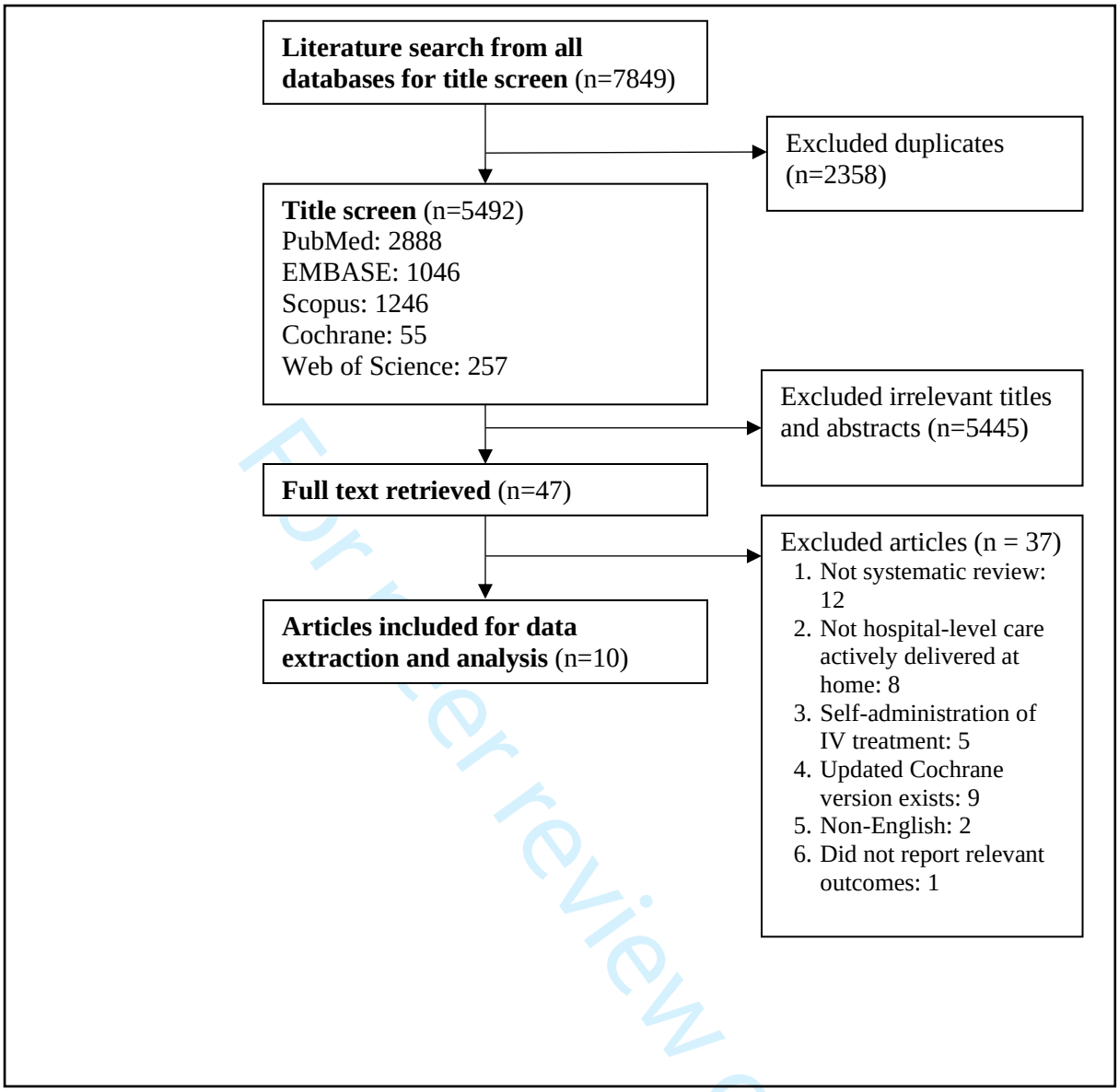


Figure 1: PRISMA flow diagram for article selection

22211

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Hospital at Home systematic review of reviews

Supplementary Table 1: AMSTAR-2 ratings

Review	1. PICO	2. A-priori*	3. Inclusion rationale for study design	4. Search strategy[a]	5. Study selection in duplicate	6. Data extraction in duplicate	7. List of excluded studies and justification	8. Detailed description of included study	9. Assessed RoB in included studies[a]
Goncalves-Bradley et al,[21] 2017	Y	Y	N	Y	Y	Y	Y	Y	Y
Langhorne et al,[9] 2017	Y	Y	N	Y	Y	Y	Y	Y	Y
Shepperd et al,[8] 2016	Y	Y	N	Y	Y	Y	Y	Y	Y
Qaddoura et al,[25] 2015	Y	P	N	Y	Y	Y	N	Y	Y
Varney et al,[26] 2014	Y	P	Y	P	N	N	N	Y	P
Caplan et al,[27] 2012	Y	P	N	P	Y	Y	N	N	Y
Sriskandarajah et al,[31] 2018	Y	P	Y	P	N	N	N	N	P
Echevarria et al,[28] 2016	Y	P	N	P	Y	N	N	Y	Y
Jeppesen et al,[29] 2012	Y	Y	N	Y	Y	Y	Y	Y	Y
McCurdy[30] 2012	Y	P	N	P	N	N	N	Y	Y

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Hospital at Home systematic review of reviews

Review	10. Sources of funding of included studies	11. Appropriate methods for meta-analyses	12. Reviewed impact of RoB on meta-analysis	13. Accounted for RoB when discussing results[a]	14. Explained heterogeneity	15. Investigated publication bias in meta-analyses	16. Reported conflict of interest	Overall Rating[b]
Goncalves-Bradley et al,[21] 2017	Y	Y	Y	Y	Y	Y	Y	High
Langhorne et al,[9] 2017	N	Y	Y	Y	Y	Y	Y	High
Shepperd et al,[8] 2016	N	Y	Y	Y	Y	Y	Y	High
Qaddoura et al,[25] 2015	Y	Y	N	Y	Y	Y	Y	Moderate
Varney et al,[26] 2014	N	NA	N	Y	Y	NA	Y	Moderate
Caplan et al,[27] 2012	N	Y	N	Y	Y	Y	N	Moderate
Sriskandarajah et al,[31] 2018	N	NA	NA	Y	N	NA	Y	Low
Echevarria et al,[28] 2016	N	Y	Y	Y	Y	N	Y	Moderate
Jeppesen et al,[29] 2012	N	Y	Y	Y	Y	N	Y	High
McCurdy[30] 2012	N	Y	N	Y	Y	N	Y	Moderate

[a]Critical item.

Abbreviations: Y = Yes, P = Partial, N = No, RoB = Risk of Bias.

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PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5-6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

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PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Nil
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Nil
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9-10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	12-14
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Nil
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Nil
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Nil
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-20
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

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Comparison of Hospital at Home Models: A Systematic Review of Reviews

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COMPARISON OF HOSPITAL AT HOME MODELS: A SYSTEMATIC REVIEW OF REVIEWS

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ABSTRACT

Objectives: To provide an overview of the safety and effectiveness of Hospital-at-Home (HaH) according to programme type (early-supported discharge (ESD) versus admission avoidance (AA)), and identify the model with higher evidence for addressing clinical, length of stay (LOS) and cost outcomes.

Methods: A systematic review of reviews was conducted by performing a search on PubMed, EMBASE, Cochrane Database of Systematic Reviews, Web of Science and Scopus (January 2005 to June 2020) for English-language systematic reviews evaluating HaH. Data on primary outcomes (mortality, readmissions, costs, LOS), secondary outcomes (patient/caregiver outcomes), and process indicators were extracted. Quality of the reviews was assessed using AMSTAR-2. There was no registered protocol.

Results: Ten systematic reviews were identified (four high quality, five moderate quality, and one low quality). The reviews were classified according to three use cases. ESD reviews generally revealed comparable mortality (RR 0.92 to 1.03) and readmissions (RR 1.09 to 1.25) to inpatient care, shorter hospital LOS (MD -6.76 to -4.44 days), and unclear findings for costs. AA reviews observed a trend towards lower mortality (RR 0.77, 95% CI 0.54 to 1.09) and costs, and comparable or lower readmissions (RR 0.68 to 0.98). Among reviews including both programme types (ESD/AA), chronic obstructive pulmonary disease reviews revealed lower mortality (RR 0.65 to 0.68) and post-HaH readmissions (RR 0.74 to 0.76) but unclear findings for resource use.

Conclusion: For suitable patients, HaH generally results in similar or improved clinical outcomes compared to inpatient treatment, and warrants greater attention in health systems facing capacity constraints and rising costs. Preliminary comparisons suggest prioritisation of AA models over ESD due to potential benefits in costs and clinical outcomes. Nonetheless, future research should clarify costs of HaH programmes given the current low-quality evidence, as well as address evidence gaps pertaining to caregiver outcomes and adverse events under HaH care.

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Strengths and limitations of this study

- The first systematic review of reviews providing a comprehensive overview of the safety and efficacy of HaH according to programme type
- Compares two major HaH programme types to offer relevant recommendations for health systems facing capacity constraints and rising costs
- Highlights evidence gaps pertaining to outcomes and process indicators in the current HaH literature
- Strength of evidence for comparison between HaH models is low due to heterogeneity in implementation and patient groups across studies

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1. INTRODUCTION

Healthcare systems around the world are grappling with ageing populations, increasing healthcare needs and expenditures, and shortages of hospital beds and manpower[1 2]. Such capacity constraints decrease the efficiency and quality of care delivery, and can lead to delays in elective and emergency treatment[3], increased mortality risk[4 5], lower patient and staff satisfaction, and higher costs for both patients and the health service[6]. Chronic morbidity in ageing populations have further driven the rise in healthcare costs[7]. These complex challenges pose a threat to the sustainability and quality of care and underscore the need to explore innovative solutions.

An alternative model of care that has been identified as a possible solution is the Hospital at Home (HaH) model, in which healthcare professionals deliver hospital-level care in patients' homes for a limited period of time[8]. There are two main types of HaH programmes— early supported discharge (ESD) and admission avoidance (AA). ESD aims to accelerate the discharge of admitted patients, thus partially substituting hospital care. AA directly admits patients into HaH based on GP referrals— thereby avoiding physical contact with the hospital, or through direct admissions from the emergency room without inpatient stay.

One of the initial impetuses behind the HaH model was to reduce adverse events associated with the hospitalisation of older patients[9]. Individual studies have demonstrated the benefits of HaH programmes in terms of safety[9 10], effectiveness[11 12], reduced iatrogenic complications[13] and cost savings[14 15]. While several systematic reviews have also been conducted, evidence for some outcomes remains unclear or conflicting, partly due to different approaches between reviews. Greater clarity may be attained by consolidating the reviews[16].

In addition, despite past suggestions that outcomes such as costs and clinical complications vary between ESD and AA models due to differing extents of inpatient care substitution[17-19], these have not been comprehensively assessed. A 2017 NICE Guideline[17] presented analyses according to programme type, but the further stratification of results based on level of care precluded any clear conclusions.

Hence, the objectives of this review were to synthesise the evidence for safety and effectiveness of HaH according to programme type, and assess which model had higher evidence for

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addressing clinical, length of stay (LOS) and cost outcomes. This would guide prioritisation in service planning, particularly for health systems that are grappling with capacity constraints and increasing costs. Process indicators and facilitators and barriers to implementation were also summarised. These are pertinent for potential adopter organisations to evaluate the feasibility of HaH in their respective settings as past studies have pointed to constraints in the HaH model (e.g. in terms of patient eligibility due to medical condition severity[20] and caregiver availability[21]). Moreover, process indicators are valuable in assessing the fidelity of implementation and identifying components that increase the success of HaH implementation[22].

2. METHODS

2.1 Data Sources and Searches

A search was performed on 14 January 2020 using PubMed, EMBASE, Cochrane Database of Systematic Reviews, Web of Science and Scopus. Search terms and criteria for inclusion/exclusion were determined a priori by all authors. Test searches were carried out and results were reviewed and refined by two authors before finalising the search strategy. The search was restricted to systematic reviews published after January 2005. While HaH pilot studies have been implemented before that, the care model took time to mature and was largely inconsistently defined prior to a seminal paper in 2005[9]. Hence, reviews after 2005 were considered more representative of the HaH literature. The search strategy employed terms that operationalised "review" and were related to "Hospital at Home" (Table 1). A pre-specified update of the search was performed on 8 June 2020. There was no registered protocol for this study.

Table 1: Search syntax for database search

Database	Number	Search
PubMed	#1	Search: ((((((home care services, hospital-based [mesh]) OR ("hospital at home")) OR ("home hospital*")) OR ("early supported discharge" [Title/Abstract])) OR ("admission avoidance" [Title/Abstract])) OR (((home[Title/Abstract]) AND hospital[Title/Abstract]))) AND ((((review[Title/Abstract] OR overview[Title/Abstract])) OR meta-analy*[Title/Abstract]) OR meta-review*[Title/Abstract]) Filters: English, from 2005 – 2020
Scopus	#1	((TITLE-ABS-KEY (hospital* W/2 home)) OR (ALL ("home hospital")) OR (ALL ("hospital at home")) OR (TITLE-ABS-KEY (early W/1 support* W/1 discharge)) OR (TITLE-ABS-KEY (admission W/1 avoid*))) AND ((TITLE-ABS-KEY (overview OR review) OR TITLE-ABS-KEY (meta AND analy*) OR TITLE-ABS-KEY ("meta review") OR TITLE-ABS-KEY ("review of reviews"))) AND PUBYEAR > 2004 AND (LIMIT-TO

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		(SUBJAREA , "MEDI") OR LIMIT-TO (SUBJAREA , "NURS") OR LIMIT-TO (SUBJAREA , "SOCI") OR LIMIT-TO (SUBJAREA , "HEAL")) AND (LIMIT-TO (LANGUAGE , "English")
Cochrane	#1	MeSH descriptor: [Home Care Services, Hospital-Based] explode all trees
	#2	MeSH descriptor: [Home Care Services] explode all trees
	#3	MeSH descriptor: [Hospitalization] explode all trees
	#4	#2 and #3
	#5	(early NEAR/1 support* NEAR/1 discharge):ab,ti
	#6	(admission NEAR/1 avoid*):ab,ti
	#7	(hospital* NEAR/3 home):ab,ti,kw
	#8	"hospital at home"
	#9	"home hospital*"
	#10	#1 or #4 or #5 or #6 or #7 or #8 or #9 with Cochrane Library publication date from Jan 2005 to Feb 2020, in Cochrane Reviews
Web of Science	#1	ts= (hospital* NEAR/3 home)
	#2	ts = "early support* discharge"
	#3	ts = "admission avoid*"
	#4	ts = "home hospital*"
	#5	Ts = "hospital at home"
	#6	#5 OR #4 OR #3 OR #2 OR #1
	#7	ts= (overview OR review OR "meta analy*" OR "meta review" OR "review of reviews")
	#8	#7 AND #6
	#9	ESCI Timespan=2005-2020
EMBASE	#1	'home care'/exp
	#2	'hospital'/exp
	#3	#1 AND #2
	#4	(hospital* NEAR/3 home):ab,ti,kw
	#5	'early support* discharge':ab,ti,kw
	#6	'admission avoid*':ab,ti,kw
	#7	#3 OR #4 OR #5 OR #6
	#8	overview:ab,ti OR review:ab,ti OR 'meta analy*':ab,ti OR 'meta review':ab,ti OR 'review of reviews':ab,ti
	#9	#7 AND #8
	#10	#7 AND #8 AND [english]/lim AND [2005-2020]/py

2.2 Study Selection

We included systematic reviews of HaH interventions, with HaH defined as the active provision of care by healthcare professionals in patients' homes for acute and post-acute conditions that otherwise would require hospital inpatient care. While HaH is commonly defined as the provision of acute care at home, we aligned our operational definition with Cochrane reviews which have included trials providing both acute and post-acute services[23 24], or post-acute care alone[25-27]. In addition, care services should only be provided for a time-limited period[8 28]. Systematic reviews including other interventions were eligible if subgroup analyses for HaH were presented. We included reviews which reported at least one of the following: clinical outcomes, resource use or process

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indicators. Process indicators were defined as measures of the processes involved in achieving programme objectives[29]. English-language systematic reviews of all study designs were eligible.

Reviews of interventions that were delivered in outpatient settings, provided long-term care, or primarily involved patient self-care at home were excluded. We excluded reviews of palliative, obstetric, paediatric and mental health HaH schemes due to differences in these groups' care needs and objectives from general medicine patients. We excluded reviews that were entirely overlapped by a subsequent update or review of higher quality, as determined using the Assessment of Multiple Systematic Reviews- 2 (AMSTAR-2)[30]

Literature search and shortlist of the articles were done consistently by one investigator, and shortlisted articles were reviewed by two authors for final inclusion.

2.3 Data Extraction and Quality Assessment

Review characteristics and relevant outcomes were extracted from the reported findings and tables in the reviews. Primary outcomes of interest were mortality, readmissions, costs, and LOS. As the entire treatment duration under HaH is considered inpatient stay, LOS in ESD schemes includes both the stay in the hospital and home. For AA schemes, LOS refers to LOS in the home before discharge from HaH. Secondary outcomes of interest were patient and caregiver satisfaction and caregiver burden. Process indicators (recruitment rates, adverse events) and facilitators and barriers to implementation were also extracted.

Methodological quality of the reviews was assessed using AMSTAR-2[30] which provides overall ratings (high, moderate, low, critically low) based on weaknesses in critical domains. In line with Cochrane guidelines[31], methodological quality of the primary studies was determined using the review authors' original assessments of risk of bias.

2.4 Data Synthesis and Analysis

Findings from included reviews were grouped, narratively summarised and compared. Due to heterogeneity between studies, a meta-analysis was not possible.

2.5 Patient Involvement

Patients or the public were not involved in this research. Ethics review was not sought and patient consent was not required.

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3. RESULTS

The search returned 7869 potentially relevant references. No relevant papers were identified from the search update. Ten reviews were eventually included. The PRISMA flow diagram in Figure 1 details the article selection process.

3.1 Review Characteristics

The ten reviews (four Cochrane reviews, six non-Cochrane reviews) were published between 2012 and 2017. Cumulatively, there were 100 relevant primary studies, with substantial overlap in studies between reviews (40%). The studies were published between 1976 and 2016, and were mostly from the UK, US, Australia, Italy and Spain.

Some reviews either assessed only ESD or AA interventions, while several reviews included both ESD and AA studies. The reviews were categorised according to three use cases: ESD[28 32], AA[8 33 34] and ESD/AA[35-39]. Characteristics of the reviews are summarized in Table 2. Service structure of the HaH programmes in the primary studies varied both within and between reviews. There was heterogeneity in the team coordinating care (e.g. hospital versus community-based), care team composition (e.g. nurses, physiotherapists, occupational therapists), programme components (e.g. additional elements such as patient and caregiver education), number of home visits, availability of after-hours support, and specific medical services provided (e.g. home oxygen, IV fluids).

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Table 2: Characteristics of included reviews

S/N	Review	Date assessed as up-to-date	Study objectives	Participant characteristics	No. of relevant studies[a]	Study design (No. of studies)	Risk of bias of primary studies	AMSTAR-2 rating
Early Supported Discharge								
1	Goncalves-Bradley et al,[28] 2017	Jan 2017	To determine the effectiveness and cost of HaH compared with inpatient hospital care.	Elderly patients with a mix of medical conditions (e.g. cardio-respiratory diseases, neurological conditions, fractures), stroke patients, and elective surgical patients (e.g. hip and knee replacements)	32	RCT (32)	Majority of the trials were at low risk of bias.	High
2	Langhorne et al,[32] 2017	Apr 2017	To assess the effect of ESD on hospital LOS, patient and carer outcomes and resource use in comparison with conventional hospital care.	Stroke patients	15	RCT (15)	Trials were generally at low risk of bias, but blinding was impossible.	High
Admission Avoidance								
4	Shepperd et al,[8] 2016	Mar 2016	To determine the effectiveness and cost of managing patients with admission avoidance HaH compared with inpatient hospital care.	Patients with COPD, stroke, cellulitis, community-acquired pneumonia, fever, neutropenia, dementia, neuromuscular disease and other acute medical conditions	16	RCT (16)	Majority of the studies were at low risk of bias, but unclear risk of bias for selective reporting and performance bias.	High
5	Qaddoura et al,[33] 2015	Oct 2014	To assess the effect of HaH on clinical, patient-centred, and cost outcomes in heart failure in comparison to hospital care.	Patients with heart failure	5	RCT (3), before-after study (2)	RCTs were modest in quality. Blinding was not possible for all studies. Several studies had unclear risk of selection bias, high risk of attrition bias, and lack of protocol standardisation. Lack of control for confounders in observational studies.	Moderate

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6	Varney et al,[34] 2014	2013	To assess the efficacy of AA HaH services that admit patients directly from the ED in comparison to hospital care.	Patients with COPD, cellulitis, pneumonia, heart failure, dementia with acute illness, deep vein thrombosis and other conditions.	16	RCT (9), case-control study (1), observational (5), cost-analysis (1)	Majority of the studies were at low risk of bias.	Low
Early Supported Discharge and Admission Avoidance								
7	Caplan et al,[35] 2012	Feb 2012	To assess the effect of HaH services that significantly substitute for in-hospital time (≥ 7 days or ≥ 25% of LOS of inpatient controls) on mortality, readmissions, patient and carer satisfaction, and costs.	Medical, surgical and rehabilitation patients >16 years (conditions not specified)	56	RCT (56)	Almost all studies were not blinded.	Low
8	Sriskandarajah et al,[39] 2018	Sep 2016	To review observational studies evaluating the safety and effectiveness of HaH and OPAT.	Patients with a mix of conditions (conditions not specified)	6	Observational (6)	Risk of bias ranged from low to high; majority of studies had moderate risk of bias.	Low
7	Echevarria et al,[36] 2016	Nov 2014	To assess the safety, efficacy and cost of ESD and AA compared to hospital care for patients with acute exacerbation of COPD	Patients with acute exacerbation of COPD	8	RCT (8)	Low risk of selection bias; high risk of performance bias on subjective outcomes as blinding was not possible; risk of attrition bias in 6 trials.	Moderate
8	Jeppesen et al,[37] 2012	Oct 2010	To evaluate the efficacy of HaH compared to hospital inpatient care in acute exacerbations of COPD.	Patients with acute exacerbation of COPD	8	RCT (8)	Majority of trials had low risk of bias; high risk of bias in 1 trial due to selective reporting and "other biases" (not described).	High
9	McCurdy[38] 2012	Aug 2010	To compare HaH with inpatient hospital care for patients with acute exacerbations of COPD who present to the emergency department (ED).	Patients with acute exacerbation of COPD	6	RCT (6)	Majority of the studies had high risk of bias due to lack of allocation concealment, unclear methods for randomisation, unclear blinding of assessors, inadequate sample sizes and improper ITT analyses (withdrawals/dropouts ignored).	Moderate

[a] Primary studies that did not fulfil eligibility criteria were excluded from this count.

Abbreviations: LOS = Length of stay. COPD = chronic obstructive pulmonary disease. OPAT = outpatient parenteral antimicrobial therapy.

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3.2 Quality of Studies

There were four reviews[8 28 32 37] of high quality, five[33-36 38] of moderate quality, and one [39] of low quality. Main areas of weakness were the lack of a registered protocol and failure to provide a list of excluded studies (refer to Supplement Table 1 for AMSTAR-2 ratings).

The randomised trials were mostly assessed to be at low risk of bias, while the observational studies were of moderate to high risk of bias.

3.3 Outcomes

Detailed findings and summary measures are reported in Table 3. As a meta-analysis was not conducted, effect estimates across reviews were summarised descriptively in the main text where feasible to do so (i.e. when the same type of effect measure was used across summarised reviews). Confidence intervals were included if the effect measure was only reported by one review.

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Table 3: Detailed primary outcomes in ESD, AA and ESD/AA

Review	Mortality	Readmissions	LOS	Costs and cost-effectiveness
ESD				
Goncalves-Bradley et al,[28] 2017	Stroke patients: RR 0.92 at 3 to 6 months (95% CI 0.57 to 1.48, P = 0.71; 11 trials; moderate-quality evidence). Mix of conditions: RR 1.07 at 3 to 6 months (95% CI 0.76 to 1.49, P = 0.71; 8 trials; moderate-quality evidence). COPD patients: RR 0.53 (95% CI 0.25 to 1.12, P = 0.096; 5 trials; low-quality evidence). Elective surgery patients: little or no difference (range of -3.8% to 0.1%; 3 trials; low-quality evidence).	Stroke patients: RR 1.09 (95% CI 0.71 to 1.66, P = 0.70; 5 trials; low-quality evidence). Mix of conditions: RR 1.25 at 3 months (95% CI 0.98 to 1.58, P = 0.069; 9 trials; moderate-quality evidence). COPD patients: RR 0.86 (95% CI 0.66 to 1.13, P = 0.29; 5 trials; low-quality evidence). Elective surgery patients: little or no difference (5 trials; low-quality evidence)	*Hospital LOS* Stroke patients: MD -6.68 days (95% CI -10.19 to -3.17, P < 0.001; 4 trials; moderate-quality evidence). Median reductions of 8 to 15 days in 6 other trials. Mix of conditions: MD -0.36 to -22 days (8 trials). Pooled data from 4 trials: MD -6.76 days (95% CI -10.60 to -2.92, P < 0.001; moderate-quality evidence). Orthopaedic surgery patients: MD -4.44 days (95% CI -6.37 to -2.51, P < 0.00001; 4 trials; moderate-quality evidence). *Total LOS (with home LOS)* Mix of conditions: MD 6.43 days (95% CI 2.84 to 10.03, P < 0.001; 3 trials). Elective surgery patients: MD 2.79 days (95% CI 0.77 to 4.81; 2 trials).	Initial and total costs were similar, lower or higher than inpatient care across trials (low to very low-quality evidence). Only one trial factored in community costs; mean hospital cost savings (at 6 months post-randomisation) were reduced from AUD 4678 (95% CI AUD 2676 to AUD 6680) to AUD 2013 (95% CI AUD 669 to AUD 4696) upon factoring in community costs, primarily due to the costs of home-based rehabilitation.
Langhorne et al,[32] 2017	OR 1.04 at 6 months (95% CI 0.77 to 1.40, P = 0.81; 16 studies; moderate-quality evidence). Reduced odds of death or dependency in the moderate stroke subgroup (OR 0.77, 95% CI 0.61 to 0.98) than the severe stroke subgroup (OR 1.40, 95% CI 0.83 to 2.36); P = 0.04.	OR 1.09 at 6 months (95% CI 0.79 to 1.51, P = 0.59; 6 trials; low-quality evidence).	MD -5.5 days (95% CI -2.9 to -8.2, P < 0.0001; 16 trials; moderate-quality evidence). Larger reduction for severe stroke subgroup (MD -28 days, 95% CI -17 to -40) than moderate severity subgroup (MD -3 days, 95% CI -1 to -7), P < 0.0001.	Total costs ranged from 23% less to 15% greater than inpatient care (7 trials).
AA				

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Shepperd et al,[8] 2016	RR 0.77 at 3 months (95% CI 0.54 to 1.09, P = 0.15; 5 trials; moderate-quality evidence). RR 0.77 at 6 months (95% CI 0.60 to 0.99, P = 0.04; 6 trials; moderate-quality evidence).	RR 0.98 at 3 to 12 months (95% CI 0.77 to 1.23, P = 0.84; 7 trials; moderate-quality evidence).	MD -8.09 to 15.90 days (7 trials)	Trend towards lower initial and total costs (8 trials). Estimates (boot-strapped MD) of costs varied from GBP -210.90 (95% CI GBP -1025 to GBP 635.47) at 3 months, to GBP -304.72 (95% CI GBP -1112.35 to GBP 447.89) per episode. In 1 trial, cost differences were no longer significant upon including the costs of informal care, although HaH might still be cheaper (MD: GBP -2216, 95% CI GBP -4771 to GBP 339).
Qaddoura et al,[33] 2015	RR 0.94 (95% CI 0.67 to 1.32); P = 0.18; 3 trials). Significant reduction in 1 observational study (P < 0.05).	RR 0.68 (95% CI 0.42 to 1.09, P = 0.34; 2 trials); non-significant reduction in 1 trial; significant reductions in 2 observational studies (MD -3.80 to -0.70 readmissions per patient). Similar total readmission rates (including readmissions during HaH care) to inpatient care in 1 study.	Significantly longer LOS in 3 studies (MD 3.0 to 9.1 days)	Significant reductions in costs per episode (mean/median reduction €295.97 to €2691; 3 trials); trend towards lower total costs at 12 months (mean/median reduction €3070 to €3125; 2 trials).
Varney et al,[34] 2014	No significant differences across 5 trials.	Low rates of unplanned hospital admissions across 4 observational studies.	Longer LOS across 3 trials, with significant effects in 2 trials.	Lower costs across 6 studies, with significant effects in 4 studies; statistical significance not reported in the other 2 studies.
ESD/AA				
Caplan et al,[35] 2012	Overall: OR 0.81 (95% CI 0.69 to 0.95, P =0.008; 42 studies) Medical patients: OR 0.79 (95% CI 0.65 to 0.97, P = 0.02; 23 studies). Surgical patients: OR 0.78 (95% CI 0.29 to 2.10, P = 0.62; 5 studies). Rehabilitation patients: OR 0.83 (95% CI 0.63 to 1.08, P = 0.17; 13 studies).	Overall: OR 0.75 (95% CI 0.59 to 0.95, P = 0.02; 41 studies; $\chi^2 = 73.27$, P = 0.001). Medical patients: OR 0.76 (95% CI 0.60 to 0.97, P = 0.02; 18 studies). Surgical patients: OR 0.66 (95% C, 0.36 to 1.22, P = 0.18; 10 studies). Rehabilitation patients: OR 0.96 (95% CI 0.70 to 1.31, P = 0.79; 9 studies).	NR	Proportional differences in costs revealed lower costs under HaH. Overall: MD -1567.11 (95% CI -2069.53 to -1064.69, P < 0.001; χ^2 = 237.45, P < 0.001). Medical patients: MD -1821.69 (95% CI -2591.89 to -1051.49, P < 0.001; 5 studies). Surgical patients: MD -875.89, 95% (CI -3510.96 to 1759.19, P = 0.21; 2 studies). Rehabilitation patients: MD -2704.18 (95% CI -3973.29 to -1435.07, P < 0.001; 4 studies).

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Sriskandarajah et al,[39] 2018	NR	Readmission rates during HaH care ranged from 4.2% to 9.7% (5 studies).	NR	NR
Echevarria et al,[36] 2016	RR 0.66 at 2 to 6 months (95% CI 0.40 to 1.09, P = 0.10; 7 trials).	Post-HaH discharge readmissions at 2 to 6 months: RR 0.74 (95% CI 0.60 to 0.90, P = 0.003; 7 trials). Total readmissions: RR 0.84 (95% CI 0.69 to 1.01, P = 0.07; 7 trials).	NR	Lower costs per episode (4 trials), but 1 trial reported higher costs when societal costs (over 3 months) were factored in (HaH: €6,304; Control: €5,395; MD = 880; 95% CI -€580 to €2268), and a savings per QALY lost of €31,111.
Jeppesen et al,[37] 2012	RR 0.65 at 2 to 6 months (95% CI 0.40 to 1.04, P = 0.07; 7 trials; moderate-quality evidence).	Post-HaH discharge readmissions at 1 to 6 months: RR 0.76, (95% CI 0.59 to 0.99, P = 0.04; 8 trials; moderate-quality evidence).	NR	Trend towards reductions in direct costs (3 trials; very low-quality evidence).
McCurdy[38] 2012	Overall: RR 0.68 at 2 to 6 months (95% CI 0.41 to 1.12, P = 0.13; 6 trials). ESD: RR 0.33 (95% CI 0.13 to 0.85, P = 0.02; 3 trials). AA: RR 0.85 (95% CI 0.45 to 1.62, P = 0.63; 2 studies)	Total readmissions: RR 0.90 (95% CI 0.70 to 1.16; P = 0.41; 6 trials). Readmissions during HaH period accounted for 13% to 50% of total readmissions.	Total LOS was similar to inpatient care in 2 trials (MD 0.15 to 1.5 days), shorter in 1 trial (MD -2.9 days), and longer in 2 trials (mean/median difference 2 to 4.5 days).	NR

All risk ratios (RR), odds ratios (OR) and mean differences (MD) are for HaH compared with inpatient care.
Abbreviations: LOS = Length of stay. NR = Not reported.

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3.3.1 ESD

Clinical Outcomes

Two reviews[28 32] reported that ESD did not significantly affect the mortality rates of elderly patients with a mix of conditions (RR 1.03, 95% CI 0.76 to 1.49; moderate-quality evidence), stroke patients (moderate-quality evidence), and elective surgical patients (MD -3.8% to 0.1%; low-quality evidence).

One review[28] revealed a trend towards higher readmissions in ESD schemes for a mix of conditions (RR 1.25, 95% CI 0.98 to 1.58; moderate-quality evidence), whereas readmissions did not differ significantly between ESD and inpatient groups for stroke (RR 1.09, 95% CI 0.71 to 1.66; low-quality evidence) and surgical patients (low-quality evidence). It was unclear if these analyses included readmissions during ESD care.

Resource Use and Costs

Two reviews[28 32] concluded that ESD significantly reduced hospital LOS for all patient groups (MD -6.76 to -4.44 days; moderate-quality evidence), although high heterogeneity in results was observed. Goncalves et al[28] observed a significant increase in total LOS (in the hospital and home) across three trials for a mix of conditions (MD 6.43 days, 95% CI 2.84 to 10.03).

Two reviews[28 32] found conflicting findings for the effect of ESD on costs (very low-quality evidence). High heterogeneity in cost estimates was also observed due to differences in intervention, cost components, and cost measurements across trials. Only one trial factored in community costs; mean hospital cost savings under ESD (at six months post-randomisation) were reduced from AUD 4678 (95% CI AUD 2676 to AUD 6680) to AUD 2013 (95% CI AUD 669 to AUD 4696) upon factoring in those costs.

Patient and Caregiver Outcomes

Two reviews[28 32] reported that ESD generally led to similar or improved levels of patient and caregiver satisfaction and caregiver strain when compared to inpatient care (low-quality evidence). However, for surgical patients, three out of five trials observed lower caregiver satisfaction under ESD[28].

Process indicators

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Recruitment rates were low for all patient groups, with a median of 20% to 33% of patients meeting eligibility criteria[28 32]. Regarding adverse events and clinical complications, one review[28] reported a lack of difference between ESD and inpatient groups in three trials for surgical patients, but significantly lower rates of delirium in the ESD group in a trial for a mix of conditions.

3.3.2 AA

Clinical Outcomes

AA may reduce mortality rates compared to inpatient care. A Cochrane review[8] for a mix of conditions reported a trend towards lower mortality under AA schemes at three months (RR 0.77, 95% CI 0.54 to 1.09; moderate-quality evidence), and a significant reduction at six months (RR 0.77, 95% CI 0.60 to 0.99; moderate-quality evidence). Two other reviews[33 34] did not find significant effects on mortality (RR 0.94; 95% CI 0.67 to 1.32)[33], but raised the issue of under-powering.

A review for heart failure[33] observed a trend towards fewer readmissions (post-discharge from HaH) under HaH programmes (RR 0.68, 95% CI 0.42 to 1.09). One study reported total readmissions (inclusive of readmissions during HaH care), which was comparable between AA and inpatient groups. Another review[8] did not find significant differences in total readmissions between groups (RR 0.98, 95% CI 0.77 to 1.23; moderate-quality evidence).

Resource Use and Costs

Two reviews[33 34] reported longer LOS (at home) under HaH schemes (MD 3.0 to 9.1 days). Another review[8] concluded that the effect on LOS was unclear due to high heterogeneity (MD -8.09 to 17 days). However, there was a general trend towards longer LOS.

Three reviews[8 33 34] found that HaH reduced costs for the acute episode, with this effect reaching significance in several studies (low-quality evidence). With regards to the mean cost savings per episode, one review[8] reported estimated savings of GBP 304.72 found in one study (95% CI GBP -447.89 to GBP 1112.35), and another review[33] reported cost savings ranging from €295.97 to €2691. However, Shepperd et al[8] suggested that the cost benefit may be offset by the costs of informal care as one trial revealed that the cost reduction was no longer significant upon including informal care costs. Two reviews[8 33] revealed a trend towards lower total costs (inclusive of follow-up costs) under HaH, although Qaddoura et al[33] concluded that the effect was unclear as few studies included indirect costs.

For peer review only

Patient and Caregiver Outcomes

One review[8] observed significantly higher levels of patient satisfaction in HaH than inpatient care across most trials (low-quality evidence). Two other reviews[33 34] also found high levels of satisfaction among HaH patients in observational studies. For caregiver outcomes, Shepperd et al[8] concluded that there was insufficient evidence as two trials revealed conflicting results.

Process Indicators

One review[8] reported that recruitment rates for HaH were low but no data was provided. One review[34] concluded that HaH generally did not affect the rate of adverse events. Shepperd et al[8] reported a lack of difference in clinical complications in one trial, and significantly lower rates of bowel and urinary complications and behavioural problems (in dementia patients) under HaH in two trials.

3.3.3 ESD/AA

Clinical Outcomes

One review[35] for a mix of conditions reported significantly lower mortality under HaH as compared to inpatient care (OR 0.81, 95% CI 0.69 to 0.95). However, this analysis should be interpreted with caution as several of the included primary studies[40-43] did not meet the criteria of HaH (e.g. nature of care was long-term care or active care was not provided by healthcare professionals). Three reviews for COPD patients[36-38] reported a trend towards lower mortality under HaH (RR 0.65 to 0.68; moderate-quality evidence).

Caplan et al[35] reported that HaH significantly reduced readmissions compared to inpatient care (OR 0.75, 95% CI 0.59 to 0.95). However, interpretation of this result is constrained as the analysis included four irrelevant studies (psychiatric studies), and this subgroup had the largest effect size amongst the various specialties. It is also unclear if readmissions during HaH care were included in these analyses. For COPD patients, two reviews[36 37] found significant reductions in post-HaH discharge readmissions (RR 0.74 to 0.76; moderate-quality evidence), while two reviews[36 38] observed non-significant reductions in total readmissions (RR 0.84 to 0.90).

Healthcare Resources and Costs

For peer review only

One review[38] concluded that the effect of HaH on total LOS for COPD patients could not be determined due to conflicting findings.

Caplan et al[35] reported that HaH significantly reduced costs compared to inpatient care, albeit with significant heterogeneity. The cost components of these analyses were unclear. Two reviews for COPD patients[36 37] concluded that evidence of cost savings under HaH was of very low quality due to heterogeneity. Echevarria et al[36] reported that one study observed higher costs for HaH than inpatient care when societal costs were included. The study also found that HaH was associated with a savings per QALY lost of €31,111.

Patient and Caregiver Outcomes

One review[35] reported that majority of the trials revealed higher patient and caregiver satisfaction, and lower caregiver burden under HaH. Two reviews[37 38] reported that the effect of HaH on patient satisfaction for COPD patients was inconclusive due to limited data (low quality evidence). One trial observed that HaH did not affect caregiver satisfaction[37].

Process Indicators

COPD-specific reviews[36-38] reported low recruitment rates, with the percentage of eligible patients ranging from 11% to 39% across trials. 56% to 88.5% of eligible patients consented to participation[38].

McCurdy et al[38] concluded that HaH did not affect the rate of clinical complications, although there were lower rates of urinary tract infections under HaH in one trial. A review of observational studies[39] observed higher rates of complications in the HaH group than the inpatient group in one study, but the HaH group was also seven times larger. In another study, patients receiving IV antibiotics under AA were at lower risk of adverse effects than ESD patients.

3.4 Facilitators and Barriers

Facilitators to implementation include coordinated and multi-disciplinary HaH teams[8 32]. Barriers to implementation include physical distance of patients' homes to the hospital[37 38], medical condition stability[8 37 38], and level of disability[28 32]. ESD led to greater reductions in death or dependency[32] and higher cost-effectiveness in one trial[28] when limited to patients with mild or

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moderate disability. In addition, caregiver availability was an eligibility criteria in one-third to half of the studies across reviews[8 28].

4. DISCUSSION

Across ESD, AA and ESD/AA reviews, HaH was generally associated with similar or improved outcomes for mortality and readmissions, shorter hospital LOS, and high levels of patient satisfaction compared to inpatient care. While there was a trend towards increased total LOS under both ESD and AA, this is possibly due to the inclusion of post-acute care in some schemes[44 45]. In traditional hospitalisations, patients would likely have been transferred to step-down care facilities (e.g. community hospitals, skilled nursing facilities) which are not factored into LOS calculations.

In comparing ESD and AA reviews, ESD has been better studied, but AA may lead to greater benefits in terms of clinical outcomes and costs. AA programmes resulted in either comparable or lower mortality and readmission rates when compared to inpatient care. On the other hand, mortality and readmission rates under ESD were generally similar to inpatient care, but with a concerning trend of higher readmissions in schemes for a mix of conditions. The observed benefit of AA over ESD could have been driven by reductions in iatrogenic events as suggested by past research[34 35]. However, limited reporting on adverse events precluded clear conclusions. In addition, although the quality of evidence for costs was low across reviews, AA studies revealed consistent cost benefits, in contrast to inconclusive findings under ESD. These observations are aligned with propositions that AA may result in greater cost savings than ESD due to the complete substitution of hospital stay[34].

Reviews that included both ESD and AA studies were mainly COPD-specific reviews which revealed evidence for lower mortality and readmissions, albeit with unclear evidence for resource use. Among these reviews, one review[38] conducted subgroup analyses on clinical outcomes by programme type, but the small number of studies (n = 5) precluded conclusive interpretation. It should be noted that one review[35] conducted subgroup analyses by the degree of substitution of hospital stay (more than or less than 50%), but the 50% criterion employed did not adequately capture the distinction between ESD and AA. ESD/AA reviews with planned subgroup analyses for programme type should continue to be explored in future research.

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Based on the existing evidence, it is recommended to prioritise AA over ESD in HaH implementations due to potential benefits in terms of clinical outcomes, costs and the complete substitution of hospital stay under AA. Nonetheless, it should be cautioned that the strength of evidence for our recommendation is low due to the limitations of our indirect and post-hoc comparisons. This is especially so given the heterogeneity in implementation and patient groups across studies. We also acknowledge that it may not always be possible to maintain a clear distinction between ESD and AA in real-life implementations due to operational considerations.

4.2 Recommendations for Practice

The encouraging evidence for clinical outcomes, LOS and costs under HaH, especially the AA model, highlights an opportunity to tackle the bed shortages and spiralling costs faced by health systems. This is particularly pertinent to advanced economies which are grappling with ageing populations and an increasing burden of multi-morbid, complex cases. Moreover, the current COVID-19 pandemic provides greater impetus for HaH implementation. These schemes can be pivotal in decentralising care, facilitating rapid ramp-ups in bed capacity[46], and controlling nosocomial infections[47].

Notwithstanding the strengths of HaH, the inherent complexity of HaH poses implementation challenges[48] and several factors should be deliberated when planning such schemes. At a broad level, contextual factors for consideration include hospital location and payment structures. Given the low recruitment rates in HaH schemes, HaH may be more practical in urban areas with a sufficient and predictable patient load to allow for greater economies of scale [17 49 50]. Secondly, many countries where HaH has been widely implemented (e.g. Australia, UK) have single-payer systems and strong imperatives to keep medical costs low, whereas greater barriers are evident in countries such as the US where payment norms are still predominantly episodic and fee-for-service[51]. Nonetheless, with the shift towards value-based payments[52] and integrated care[53] across health systems, a more conducive environment for HaH may be emerging.

At the programme level, care enablers include the proximity of patients' homes to the hospital and suitability of home settings. With the policy shift towards bringing care closer to the community, especially in advanced economies[54], it may be timely to re-examine how future urban planning and

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housing designs can support models of care such as HaH. Care enablers identified in primary studies which were not included in the current paper deserve mention as well. These include skills-training of staff to provide care in HaH schemes[55 56] and assistive technologies such as telemedicine. Regarding the latter, a recent trial[57] demonstrated the safety and effectiveness of a virtual HaH programme that exclusively utilised telehealth during COVID-19.

Several reviews highlighted that HaH may not be suitable for patients with intensive care needs, or who do not live with caregivers. However, a possible concern with necessitating caregiver availability is the transfer of burden of care from hospitals to family caregivers. In fact, by incorporating appropriate safeguards (e.g. panic buttons, fall-detection sensors), caregiver reliance can be reduced while ensuring that patients receive quality care. Hence, moving beyond the traditional model of reliance on care providers, HaH can serve as an avenue for patient empowerment by increasing patient involvement in their own care and boosting their confidence in self-management. By extending HaH eligibility to patients without caregivers, more will stand to benefit from HaH schemes as well.

4.3 Evidence gaps

The most well-studied diagnostic groups were COPD and stroke, with evidence of positive clinical outcomes for COPD patients. Past research has indicated that other conditions are also suitable for HaH, such as cellulitis and deep-vein thrombosis[58 59]. However, there was a lack of subgroup analyses or reviews for these conditions, hence the effects of HaH for these groups require further clarification.

It was unclear in multiple reviews if early readmissions during HaH care were factored into analyses for readmissions, hence it was challenging to make comparisons across reviews. Future reviews should clearly report both early and total readmissions. The evidence on costs was of low quality due to heterogeneity across studies, and few studies assessed cost-effectiveness. Future studies could consider full economic evaluations that include all direct and indirect costs (e.g. informal care costs) to better guide policy and decision making. Further clarification on caregiver outcomes, especially caregiver burden, is also required given caregivers' crucial roles in HaH programmes[21].

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Lastly, there was limited reporting on facilitators, barriers and process indicators such as adverse events. Future trials and reviews should report these given that one of the initial objectives of developing the HaH model was to reduce iatrogenic events, and the incidence of iatrogenic events is also a frequently theorised comparison point between ESD and AA.

4.4 Limitations

The first limitation is the lack of a registered protocol for this study. In addition, the current research might have missed primary studies that were not included in the systematic reviews, and we also excluded grey literature. However, we attempted to address these limitations by highlighting relevant findings from primary studies and government publications.

Outcomes were mainly limited by small sample sizes and heterogenous measurement tools in the primary studies. Also, two reviews[32 35] included primary studies that did not meet the criteria of HaH, hence there were likely inaccuracies in the reported effect estimates.

Lastly, overlaps in primary studies across reviews could have inflated results. This limitation was most pronounced for COPD-specific reviews (90% overlap), but these reviews were included as they provided insight on different outcomes.

4. Conclusion

HaH is largely safe and effective for suitable patients requiring hospital-level care. HaH generally leads to similar or improved clinical outcomes, shorter hospital LOS, and high patient satisfaction when compared to inpatient care. HaH particularly warrants greater attention in health systems that are challenged by capacity constraints and rising costs. Moreover, based on preliminary comparisons, AA schemes should be prioritised in further explorations of HaH due to the complete substitution of hospital stay, and possible cost savings which will likely increase with scaled-up programmes. However, for both AA and ESD, several outcomes and process indicators still require further clarification, such as caregiver outcomes, cost-effectiveness and clinical complications.

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Figure 1. PRISMA flow diagram for article selection.

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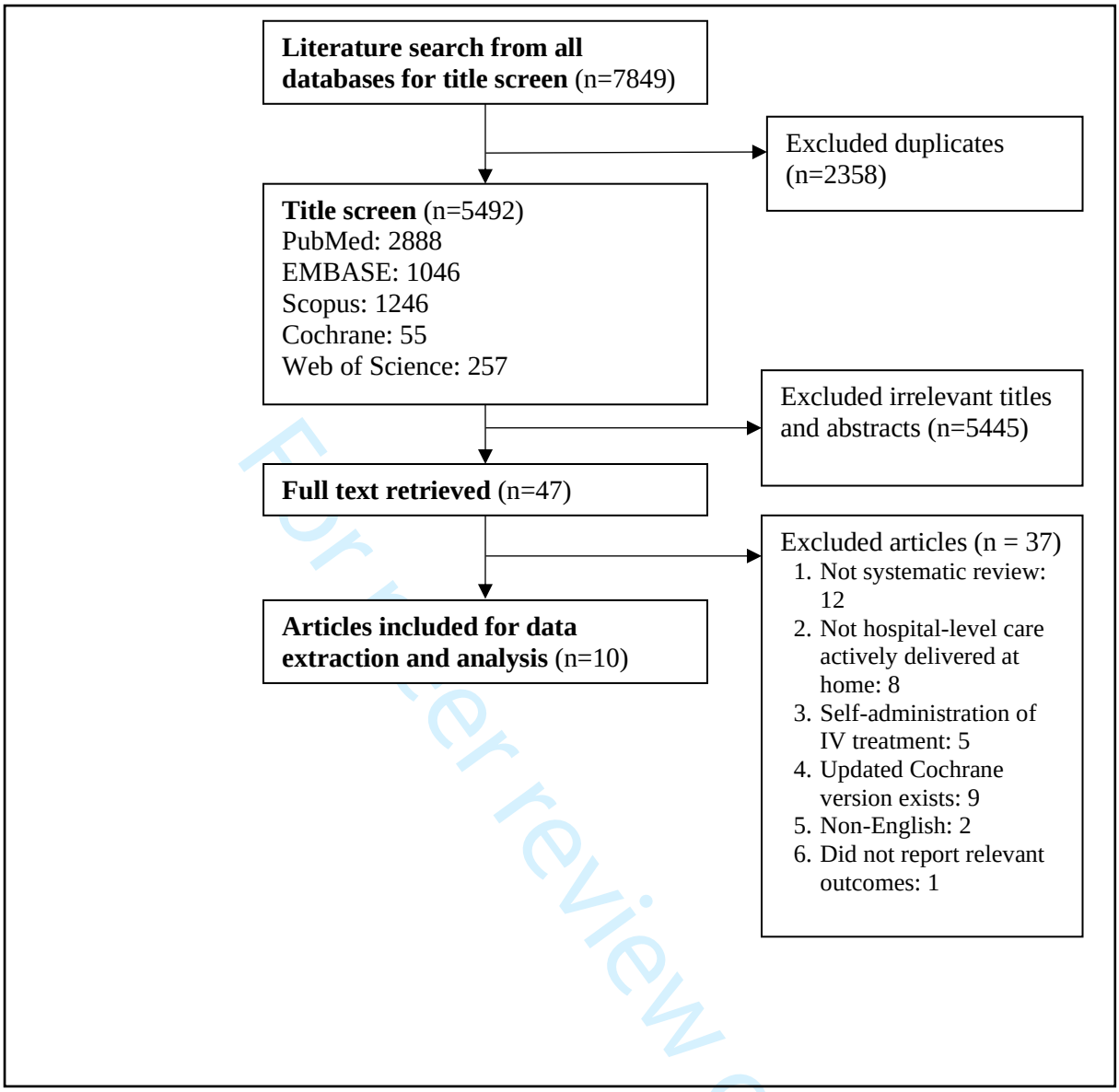


Figure 1: PRISMA flow diagram for article selection

22211

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Supplementary Table 1: AMSTAR-2 ratings

Review	1. PICO	2. A-priori*	3. Inclusion rationale for study design	4. Search strategy[a]	5. Study selection in duplicate	6. Data extraction in duplicate	7. List of excluded studies and justification	8. Detailed description of included study	9. Assessed RoB in included studies[a]
Goncalves-Bradley et al,[21] 2017	Y	Y	N	Y	Y	Y	Y	Y	Y
Langhorne et al,[9] 2017	Y	Y	N	Y	Y	Y	Y	Y	Y
Shepperd et al,[8] 2016	Y	Y	N	Y	Y	Y	Y	Y	Y
Qaddoura et al,[25] 2015	Y	P	N	Y	Y	Y	N	Y	Y
Varney et al,[26] 2014	Y	P	Y	P	N	N	N	Y	P
Caplan et al,[27] 2012	Y	P	N	P	Y	Y	N	N	Y
Sriskandarajah et al,[31] 2018	Y	P	Y	P	N	N	N	N	P
Echevarria et al,[28] 2016	Y	P	N	P	Y	N	N	Y	Y
Jeppesen et al,[29] 2012	Y	Y	N	Y	Y	Y	Y	Y	Y
McCurdy[30] 2012	Y	P	N	P	N	N	N	Y	Y

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Review	10. Sources of funding of included studies	11. Appropriate methods for meta-analyses	12. Reviewed impact of RoB on meta-analysis	13. Accounted for RoB when discussing results[a]	14. Explained heterogeneity	15. Investigated publication bias in meta-analyses	16. Reported conflict of interest	Overall Rating[b]
Goncalves-Bradley et al,[21] 2017	Y	Y	Y	Y	Y	Y	Y	High
Langhorne et al,[9] 2017	N	Y	Y	Y	Y	Y	Y	High
Shepperd et al,[8] 2016	N	Y	Y	Y	Y	Y	Y	High
Qaddoura et al,[25] 2015	Y	Y	N	Y	Y	Y	Y	Moderate
Varney et al,[26] 2014	N	NA	N	Y	Y	NA	Y	Moderate
Caplan et al,[27] 2012	N	Y	N	Y	Y	Y	N	Moderate
Sriskandarajah et al,[31] 2018	N	NA	NA	Y	N	NA	Y	Low
Echevarria et al,[28] 2016	N	Y	Y	Y	Y	N	Y	Moderate
Jeppesen et al,[29] 2012	N	Y	Y	Y	Y	N	Y	High
McCurdy[30] 2012	N	Y	N	Y	Y	N	Y	Moderate

[a]Critical item.

Abbreviations: Y = Yes, P = Partial, N = No, RoB = Risk of Bias.

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PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5-6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

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PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Nil
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Nil
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9-10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	12-14
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Nil
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Nil
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Nil
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-20
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

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